



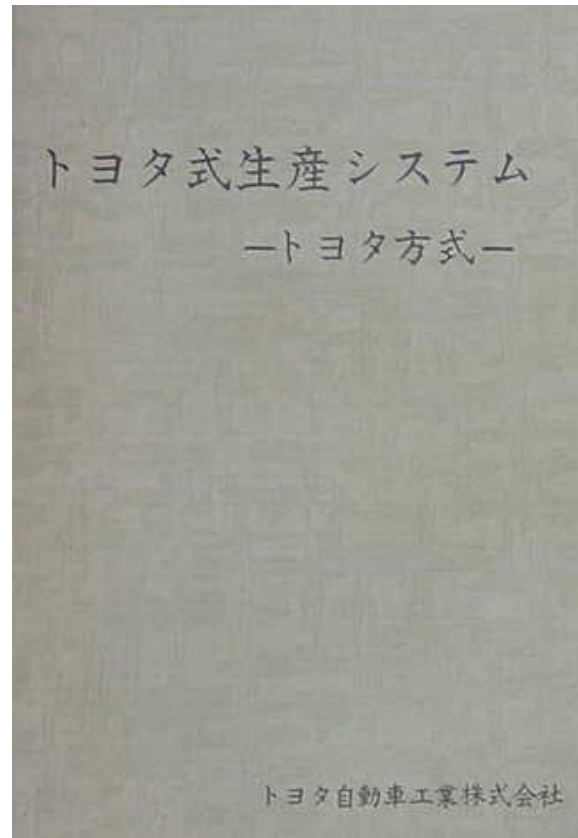
1973 TOYOTA

Production System Manual



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1973 Toyota Production System Manual



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Chapter 1

Section 1 – Toyota Cost & Labor Reduction Method

The basic ideas of the *Toyota Production System* (TPS). **TPS is a sequence of operations with an objective of improving productivity by reducing unnecessary, unreasonable and uneven activities, thus decreasing our production costs.**

Although its underlying ideas are based on general *Industrial Engineering* (IE) theories, these fundamental ideas were developed over a long period of time exclusively for the improvement of the production system of the Toyota Motor Corporation. It is not just a simple application of IE theories.

In this section, rather than merely listing facts about the fundamental ideas, we would like to review the characteristics of TPS and demonstrate a few of its concepts.

Company-wide IE Directly Connected to Management

Much has been said about production management systems; however, there is no single method of "*How to Make Things*" that would work for every production process and for all types of products.

Accordingly, some companies could make a product using a single person, another would use two people, while a company that is completely unconcerned with production methods would produce it using three people.



In the last case, the company which used three people would likely have more warehouses, vehicles, pallets, conveyors, and other equipment. This can cause an increase in the number of indirect employees and production cost could be double (or more), making a vast difference in profits (if they are profitable at all).¹

We do not have concrete statistics that clarify this relationship; however, we can imagine that such difference happens in many companies. Moreover, this difference will not only be a

¹ In 1950, Toyota realized how ineffective they really were after Eiji Toyoda visited Ford's River Rouge Plant. Toyota estimated that it took about nine employees to accomplish the same work as a single Ford employee.

matter of the present situation. Every single year, investment will be repeated in similar ways because the annual plans are designed based on the thinking that created the current situation.

Moreover, once the fixed costs (manufacturing plants, equipment, machinery, etc.) have been invested, these fixed costs are irretrievable even if they later turn out to have been unnecessary. Considering the serious risks of over investment, this is a significant issue for every company.

The only way to avoid the risk of over-investment is to continuously improve the production effectiveness in every aspect of the operation process, reducing the basic units of consumption for labor, equipment and machinery. Then reflect these continuous improvements in the production plans for the next phase.



Looking at things from this perspective, Industrial Engineering will have a major effect on business management. Companies that do not practice IE sufficiently are like splendid castles built on sand. The company may look successful during times of economic expansion, but during economic contractions, they can collapse as if hit by a typhoon or an earthquake.

The productivity improvement routines are an essential foundation of the corporate management system for Toyota. All departments that are involved in production handle things in the overall most effective way. The production system is based on the following concepts:

Leveling Production Plans.

If you think solely about the production efficiency at the final assembly plant, you may think it is most efficient for the plant to run only one product; however, this would rather result in unnecessary activity in the preceding stages of production.

Minimize Lot Size

Lots should be minimized in typical lot production processes, such as using a press. This is not just about the inventory piling up or increasing the labor for transport, but there is also a risk of experiencing a stock-out, caused by mistaken order priorities. The capacity of the processes could be misjudged as insufficient, which would then lead to an unnecessary extension of production lines (or even additional buildings).

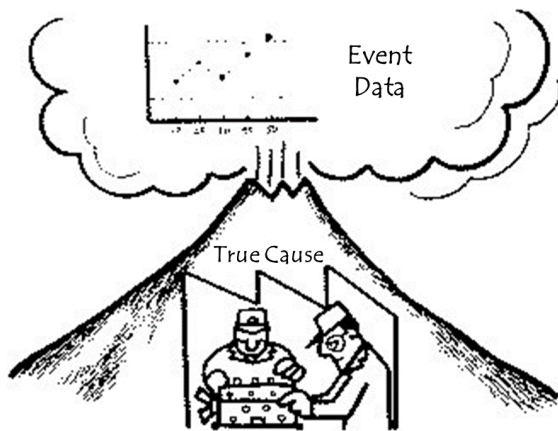
To enable small lot production without causing a reduction in your production capacity or adding costs, it is necessary to improve your changeover capability.

Just-In-Time

Produce only the required items, at the required time, and in the required amount. The aim of this is to reduce the unnecessary activity of overproduction, as well as to allow you to visualize the surplus capacity at that moment.

Scientific Attitude

In the workplace, problem solving must clearly follow the following steps: the actual situation is the starting point, investigate for root cause, and then develop a solution. In other words, when it comes to the workplace, facts are regarded as most valuable.



For example, regardless of how much data you are shown, it is difficult to grasp the actual situation of the workplace from data. If a defect occurs, finding it out from data (*reports*) would mean a delay in taking countermeasures.

Even worse, due to misunderstanding of the root cause of the failure, you may also fail to prevent the recurrence of the failure.

There is no place better than the actual workplace where you can grasp the real

situation properly of what is happening there. When there is a failure, you must contain it immediately; it is easier for you to find the true cause, and take immediate measures.

In TPS, when it comes to the workplace, we regard the real facts to be more accurate than data.

When you have a problem and your process of finding out the root cause is insufficient, the countermeasures you design will also be out of focus. That is why it is necessary to use the 5-Whys method, and ask yourself “Why, why, why...,” until you find the real root cause.

To make this method work, you should keep the following points in mind:

Make Details of the Problem Obvious

If the problem is known, it is relatively easy to implement countermeasures. The hard part is finding out what the problem is. That is why we use various tools to signal a problem.

Objective of Problem-Solving should be Clear.

Improvement is based on need. The root cause must be identified, and then solved.

Countermeasures taken without fully investigating the root cause would be merely a provisional measure, which does not lead to the prevention of reoccurrence of the problem.

Take Countermeasures for Every Single Failure

Even the failure occurs only once in a 1000, if you have facts, you must investigate its cause, and take preventive measures. It is vital to be able to spot the infrequent defects that are likely to go unnoticed.

Be Practical

There are two aspects to being practical. One approach can be defining step-by-step measures. You can set a high goal, but your practice toward this goal should be performed step-by-step. The second is to attach importance to results.

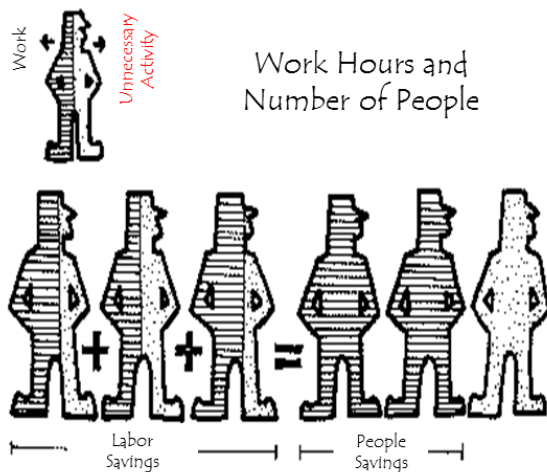
Based on these ideas, we think as follows:

Start from Work Improvement to Equipment Improvement

General sequence is; people improvement (training), then process (including layout), then equipment, then design. (More details in Section 6)

"Man-hours", "number of people", "labor cost saving" and "labor saving"

In calculations, you can say 0.1 hour, or 0.5 hour saved. However, you still need one person even for the work of 0.1 hour. Therefore, if a job for one person is reduced by 0.9 hours, this would not result in a reduction in labor costs. Only when the number of people are reduced, will true cost reduction be achieved. So, the improvement of labor must focus on reducing the actual number of people needed.²



In the case of introducing automated equipment, if you achieve 0.9 person's work worth of labor-reduction, 0.1 person's work remains (*often for keeping an eye on the machines*), then no reduction in people will be achieved despite the investment in equipment. This is sometimes referred to as labor-saving; a reduction of the number of actual working people that leads to a practical cost reduction is differentiated from "labor-saving" and is called "people saving".

² Reducing the number of people should be put into context. When Taiichi Ohno began actively developing TPS, Toyota took about nine times as many people to produce a vehicle as Ford. You will see a strong theme throughout TPS for improving the efficiency of production and they do this by counting the number of people it takes to do the job.

The Difference between "Automation" and "Autonomation"

With 'automation' a machine may run by itself, yet still need a person watching it to stop the machine when something goes wrong. When 'autonomation' is added to the machine, it has the ability to detect the same problems that the person guarding the machine would do, and automatically stopping the machine when a problem was detected. (More details in Chapter 2)

Checking is Reflection

The improvement is finished when the intended results are obtained. When you don't get the desired results, it's often because of unfinished work.

Try the improvement idea at the workplace first to see how it performs; check the results; improve the things still not working correctly, and then check again. By repeating this process, your improvements will provide favorable results.

The checking process is a step to reflect on the results of your work, rather than simply implementing ideas.

Economic Efficiency is the Criteria for Everything

The purpose is the reduction of costs. Therefore, the basis of the assessment is "*whether this is economically beneficial or not?*"

As for practical thinking, we have the following points:

The operation ratio of equipment is decided by the required production amount

This is emphasized to prevent falling into the wrong way of thinking about the sunk cost concept which suggests that you must make full use of facilities that have already been purchased, or else it will lead to loss.

For the same reason, in the case of multi-machine handling, we follow the same concept in focusing on the work of people, not the machines.

(More details are discussed in Section 2 as well as Chapter 2)

Spare Capacity Allows you to Practice

People who have fixed work hours will receive the same salary whether they have nothing to do (*pretending to work*) or practice improving the setup process when they have no work to do. (More details in Section 2)

Clear Position of Workplace

We regard the whole workplace as a single organism. The workplace is not just a group of hands that have entrusted their brains to management. Essentially, the workplace should play the main role. It's unacceptable for engineering to act as the boss for the actual workplace. What is important is to emphasize the autonomous action of the workplace.

To this purpose, the engineering division should fulfill its role by supporting the weaknesses in the actual workplace where they lack information or skills. They should do this without weakening the responsibility of the workplace.

Emphasize the Capacity of Adapting to Change

Plans may often change; what matters is having a production system that can adapt immediately when plans change. An adept workplace is capable of handling that.

Section 2 – Labor Reduction and Cost

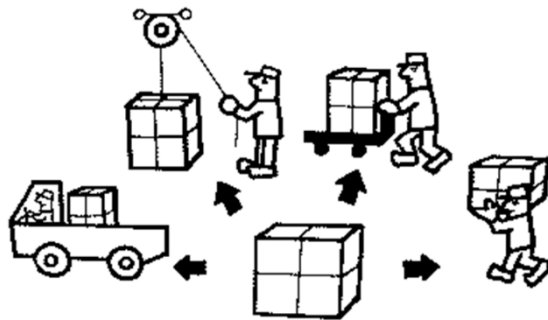
The objective of labor reduction is cost reduction.

In this section, we will talk about the relationship between labor reduction and cost reduction, while focusing on the concept of *economic efficiency*, which is the "*basic idea*" of Section 1.

Toyota's labor reduction effort is a company-wide activity concerned with the workplace, whose purpose is cost reduction. To sum it up, all the various concepts and improvement policies, must ultimately lead to cost reductions. All evaluation criteria are decided based on whether they can achieve cost reductions.

Problem of deciding which is more beneficial; A or B.

Consider the case of deciding where, if we pick option **A**, we can't pick option **B** (or possibly the other way around). In this case, we decide based on whether **A** or **B** is more beneficial for everyone in Toyota.



There are many ways to meet an objective

There might be one objective, but there are many means. For example, when we consider which is more effective: maintaining the current method of production, or doing it with one less person.

Problem of Selecting from Numerous Choices.

For "*reducing people*", there should be many approaches available. We must investigate which one among these is the most beneficial. Sometimes multiple choices occur independently; and the evaluations may occur separately, after testing the improvements, consider them relative to each other.

Sometimes there is improvement where **B** is not sufficient. If the investigation of **B** is not sufficient, this generally leads to an unreasonable improvement plan (one that costs more money.) Even though, such a plan is still valid relative to **A** and thus seemingly advantageous as a whole; it may cause rather bad results.

For example: Let's say there was a plan to reduce the number of people by one, when we installed an electric controller that costs ¥100,000. If we carry out this plan, we could reduce the number of people by one for the cost of ¥100,000, and claim that it was a big profit for Toyota.

However, after you consider the other options carefully, you may find out that one person could be reduced by changing the work procedure without investing any money. Then the plan

to improve by spending ¥100,000 cannot be considered a success, but a failure from rushed decision making.

Therefore, we don't do just anything because it's profitable. Instead we must go for the plan that we believed to be the most profitable among all the other profitable options. We must pay close attention, since this is a mistake one is easily prone to make when introducing automation.

Unnecessary Activity Increases Cost

The basic concept of the TPS is to improve the productivity by eliminating unnecessary activity. We believe it is not necessary to explain how this helps in reducing costs at this point, but when it comes to how much it helps, interpretation varies according to each person. With that in mind, we shall explain how we believe unnecessary activity affects production cost.

If we are to borrow a phrase, the definition of *“unnecessary activity in the production workplace”* would be the factors which *“only increase cost.”* For example, too many people, excessive equipment, too much inventory, and the like. Anything that exists more than needed, be it people, equipment, raw materials, or products, would only be increasing the cost. In addition, this unnecessary activity causes further unnecessary secondary activities.

For example, if there are too many people, you may create some work for them to do. The power and materials consumed for this labor can be considered as secondary unnecessary activity.

The largest unnecessary activity is caused by overstocked inventory. Let's take a closer look on this example. Suppose we have more inventory than needed. If all of it can't go in the factory alone, then there will be a need to build a warehouse, we will have to employ people to transport it to the warehouse, and we will have to buy a forklift (or truck) for each of these people. We would also need to allocate some people to the warehouses to manage the inventory and protect it. And even after doing all that, scratches, rust, or other damage are still likely to occur. Then you need more people for inspection and repair work.

Knowing what you have, and how many, is time consuming. This drives the need for considerable labor in the management department. When manual tracking reaches its limits, someone will come up with the idea of installing computers to manage the inventory.

If you don't know what you have and how many, then you will have shortage of products. And then, we will start thinking that our production capabilities must be insufficient because a stock-out occurred even though we produce it daily. So, expansion plans are placed in the capital investment for next fiscal year. And once the new equipment is introduced, we will have even more inventory to manage.

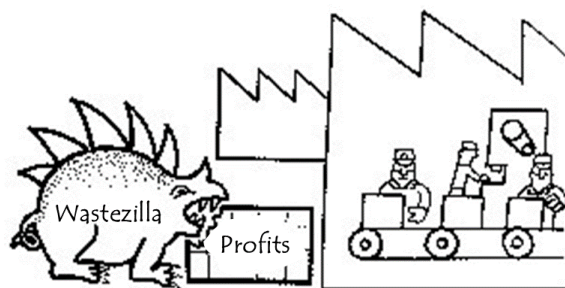
All the additional people, forklifts, pallets, manufacturing plants, computers, and equipment we mentioned here would be unnecessary. These are secondary things that were added all because of unnecessary inventory. This is all unnecessary if there were no such excess inventory.

Such an example is an assumption of the worst-case-scenario. Occurrences like this can happen anywhere at different scales. It can easily happen right away if we get a little careless. It is difficult to determine what is necessary and what is unnecessary; some details are hard to notice. Nevertheless, considering the magnitude of the impact, it is important to give close attention daily.

As shown in the earlier example, mistaking the order of priorities, ending up with a stock-out, and assuming it to be a lack of capacities, ultimately lead to investing in unnecessary equipment. A mistake that is difficult to undo.

The primary and secondary unnecessary activity described above, all will be included in the direct labor costs, depreciation costs, general and administrative expenses, thus increasing the entire production cost.

When we think this way, we cannot ignore the parts where unnecessary activity causes an increase in cost. Not to mention, a mistake like this could cause unnecessary activity to eat away the profit we make, which could compromise management itself.



This way of perceiving cost underlies the objective of TPS regarding cost reduction. We emphasize elimination of unnecessary activity, because we believe that by reducing inventory and people, clarifying the surplus capacity of the equipment, and by removing the secondary unnecessary activity, this will lead to cost reductions.

Concept of Economic Advantage

In Section One, we explained the basic idea of how we think and practice. That all the decisions are based on the question, *"Is this going to reduce costs or not?"* In the following section, let's look at the concept of the TPS from the *"economic engineering"* point of view and discuss what is economically advantageous.

With and Without Reserve Capacity

Economic advantage varies from situation to situation, if there is a surplus in production capacity or not. To put it simply, in times of surplus, we use people who have nothing productive to do, and idle machines, so there is no additional cost.

(A) The problem of in-house production or outsourcing.

There are many cases where we compare the costs of producing an article in-house or outsourcing it. If there is spare capacity of in-house production, the actual additional cost is only the variable costs such as materials and consumables. Accordingly, in-house production would be beneficial without even needing to compare.

The same can be said about the following example:

If a transport operator is waiting beside the line until the pallet is full, assigning them some line work or preparatory work will not create additional costs. This is obvious without making investigation into the profit and loss. It is a mistake to think that this labor improvement will raise your labor efficiency.

(B) The problem of making smaller lots.

If there is a surplus in general-purpose machinery such as presses, putting the problem of shortening the setup time aside, it is best to keep lots as small as possible when producing something.

If you still cannot reach the scheduled time, you use the time to the practice changeover improvements, which is more profitable than doing nothing. The reason for this is the same as (A).

If there is surplus capacity, there is no need to investigate cost, since profitability is clear as it can be. What matters here, is to always make it *obvious* when we have surplus capacity. If the surplus capacity is not obvious, this can lead to misjudgment and increase the cost.

In TPS, we take this one step further, and put effort into improvements that squeeze out the surplus. If there is surplus capacity, it may be considered as '*free*', as explained in the example above.

Sunk Costs

Costs and expense that have already been paid are irrelevant to plans or measures in the future. When you think about improvements, it is a mistake to think that this is the constraint condition.

Looking at it from the other side, we can't evaluate the effectiveness of the special-use equipment, devices, and pallets, which have been economized by the improvement activities. It is only too clear that they were once unnecessary activities.

We should take advantage of using this as learning lesson for planning future investments.

Is it really a loss to not make expensive equipment operate fully?³

In principle, once a piece of equipment is installed in the workplace, regardless of whether it is cheap or expensive, the price is irrelevant to the use of the equipment.

When you must choose between using expensive equipment **A**, or simple equipment **B**, you should choose the one whose overall operating cost is less. (Usually the expenses for **A** should be cheaper as the machine is more expensive...) However, more expensive, higher performing machinery usually has a different problem to consider; that you must put in more effort to technically master using the machine. It is noteworthy here that it is unreasonable to think that *"it is a loss to frequently change setups, because it is expensive equipment"*; there are many other conditions you should consider.

Reducing People Reduces Costs.

Index the *takt*⁴ time from the required number, and establishing processes where the people can work with 100% capacity⁵, a single person could operate multiple machines (or large variety of machines).

If there is the capacity to make more than necessary, it is wrong to think of leaving a machine idle as a loss. It is economically advantageous to set up a standard operating procedure that is centered on the work done by people, while the machine work is considered secondary.

Results of Improvement and Planning

It's not unusual to hear that after implementing improvements, a lot of fork lifts or storage racks and pallets are not used, or space has become vacant. However, this cannot be valued as an improvement, because no matter how many of them you have managed to spare, it does not create any profit.

In such cases, it's important to think about just how many unnecessary things looked as if they were necessary depending on the current production method. Give feedback to the planning department so the same type of mistake is not repeated in the next project. A plan is something that is created based on the current situation.

When there are a lot of unnecessary activities in the way things are currently done, the same methods end up being incorporated into the next plan. Furthermore, once the investments are made, they can't be undone, even if improvements are made. This is important. Such a relationship should be fully recognized. You should make efforts to eliminate the unnecessary

³ It is a trap of running expensive equipment to capacity, even if the output is not needed.

⁴ "Takt" is a German musical term for beat or rhythm. This is the frequency this is used to synchronize adjacent processes.

⁵ Loading a process to 100% capacity leads to problems like not being able to schedule time for maintenance.

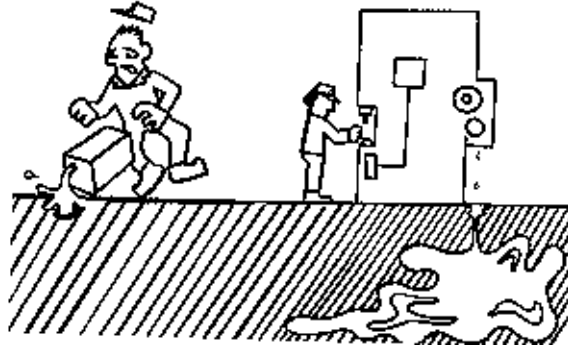
activity from sites daily, and at the same time, you should not neglect to report unnecessary activity to management once they are brought to light by the improvement process.

Continuous Costs and Temporary Costs

Costs incurred daily seem to be reasonable when you look at them individually, this makes them easy to overlook.⁶ On the contrary, temporary expenses tend to be noticed because they often have high costs, so they tend to be viewed as a loss. But if you convert the total cost of the continuous daily costs over a year into a temporary cost, it is often surprisingly expensive.

In many cases you feel the costs for the improvement are too expensive (apart from the problem of finding a cheaper way) and the continuous cost is unchecked, leading to loss.

Do not decide things just by feelings, get the facts and always do the necessary calculations. A frequently discussed issue about the installation of signal lights, and cost of setting up sign boards, or, when the improvement increased overtime.



How to use Ratios

Concept such as "*X ratio is improved*" or "*this approach more advantageous since X rate is higher*" may lead different judgment according to its purpose. There are many cases when scales such as the profit ratio of a product or an investment should not be used when choosing the beneficial product or investment. (The reason is complicated, so it has been omitted).

We would like to discuss the utilization rate in terms of cost. People often think that a decline of the utilization rate is equal to a loss. But, the most profitable way with least amount of unnecessary activity is to produce only necessary amount of what you need, when you need them. What would happen if we got obsessed over the utilization ratio and have all our machinery working at 100% capacity?

Finished products, as well as intermediate products will be piled up like a mountain, and the number of people required could be doubled or even tripled. We'll also have to purchase several times the current amount of raw materials and parts than we currently purchase. Looking at this in terms of expenditure and income, expenditure could be three or four times higher, but the income will not change at all. It is far more than just a loss.

It is best to regard the utilization rate as something to be determined by the required number of products. It is important to have the machinery available for use whenever it is needed. If

⁶ Ten minutes lost daily is equal to one week of production each year.

not, this may cause an opportunity loss and overtime which may result in a loss. TPS considers these rates separately. (See the section of Chapter 2, utilization rate and the movable rate.)

Summary

In this section, we have discussed how *cost* is considered and how a decision of *economic efficiency* is made within TPS.

Before closing this discussion, it must be noted that the economic efficiency varies in many ways according to external conditions. To put it in an extreme way, what was profitable yesterday, might be a loss today, if any new factors are introduced.

For example, when wage agreement contracts change from time-based to production-based, the problem of profit and loss emerges in a different form.

We believe that there are many other examples, as well as many challenges concerning the necessity of improvement, which have not been discussed in this section.

To sum things up, we have discussed how we should always find out the most economic option while keeping many conditions in mind.

Make this the standard of thinking for implementing improvements.

Section 3 – Labor Reduction and Quality

About Quality

In this section, we will explain how one should think of quality regarding labor reduction routines, as well as what's the relation between quality assurance activities and labor reduction. The primary subject is production quality rather than the design quality. Except for special cases, quality is referring to manufacturing quality.

Quality is Built into the Production Process

We are engaged in the automotive industry, and it is our mission to provide a trouble-free car to the customer. To do that, it is necessary to manufacture products that meet the design quality requirements, and this drives the need for quality checks. Conventionally, finished products were checked by a quality inspector, and then sent to the subsequent production process; however, no matter how much you evaluate the quality of a product that has been already manufactured, this will never lead to manufacturing products of good quality.

If the quality inspector performs a sampling-inspection and declares the products to be good, but of one them is found to be defective, if we put ourselves in our customers' place, there's no way we would say *"It can't be helped, it's only one in a thousand."*

All products are subjected to a hundred percent inspection in some form or another. This is where the elimination of the dedicated full-time quality inspectors, and the concept of *"build quality in the production process"* was born. Which means completely assuring quality in every single process step where it can only send a good product to the next process. The inspection should be done inside the process and defective products are removed on the spot.



For the concept of *"building quality into the production process,"* every single person is held responsible. The basic idea is that *"the next process is our customer,"* not sending a defective product to the next process is the basis of building quality in the production process.

Despite these efforts, sometimes a defect will be discovered in a subsequent process. If this

happens, they immediately contact the preceding process, and the department stops the production process, root cause is investigated, and countermeasures implemented.

Without prompt response, defective products will continue to be produced. In addition, it is necessary that the process that produced them must do the rework of defective product. Keeping silent just because it's a minor flaw should never happen.

Letting the next process do the rework without notifying the preceding process is unacceptable. This may result in the production of more defective products. The rework must be done by the department that produced the defective product.

Quality is the True Value of Improvement

To build products of good quality is a priority more than anything else for our manufacturing industry. No matter how many products you produce, if they are of poor quality, then customers will not buy them. Even if you reduce the production costs, if you can't sell them, there will be a loss.

In the case of automobiles, safety is especially important. We would fail our social responsibility if we bring products to the market where we *"cut corners"*, or made the excuse of *"we had our hands full"*, or *"we made it cost less"*. This can turn out to be fatal for a company.

In short, ensuring the quality is the first thing that must be considered. You are giving priority to a less important thing when you make little of it for some other reasons.

Let's consider what kind of work does the term *"quality assurance work"* refers to. Unlike in the past, when the intuition of the operator or the degree of skill played greater role, today each process is segmented, and less specialization of skills are needed. A standard operating procedure in each production process is part of quality assurance.

Standard work must be designed to ensure the required quality. If there is unevenness among the processes, checking process by visual observation or gauge must be incorporated into the standard operating procedure as a single process.

If defects are still produced under these conditions, then it is either because the standard operating procedures are not followed, or because there is some defect in machinery, equipment, molds, or tools. Let us discuss people not following the standard procedures.

Sometimes people say things like, *"When we tried reducing labor, defects increased"* or *"We reduced too many people, which shows up in our product quality."* As we explained earlier, looking at it from the concept of the TPS, this is putting the *cart before the horse*, and it should never happen.

By looking at what kind of issues take place, they can be broadly divided into the two categories:

(A) Omitting or forgetting some of the necessary procedures, under the impression that the work that must be completed in a shorter time. In other words, rather than eliminating unnecessary activity, they end up cutting corners instead.

(B) Since we previously had surplus of time allotted; creating intermediate stock and rework were possible. The labor reductions have exposed our quality defects.

The case of (A) is commonly encountered in assembly work done on conveyor lines. This mistake occurs because the line was not stopped to avoid delaying the production process

In the labor reduction activities, people are well taught about the importance of stopping the line. **The first thing to teach a new person is how to stop the line.**

By stopping the line, we can find out about the imbalance between the amount of work assigned to each person, and discover facts for the elimination of unnecessary activity. It becomes possible to solve the fundamental cause of the delay.

If people omit some work because there is not sufficient time allotted, it may be because they think the line must never be stopped. The supervisor is responsible to make sure that people understand that passing a complete product to the next process is more important, even if it requires stopping the line.

There is no need to obsess over line speed or *Takt* time. It's important to clarify that *"Takt time and the number of people are unrelated."* The person is doing all that is required at their own pace, so a cycle of work is completed.

If they don't finish within the Takt time, the line may be stopped until it is completed. Making sure that work fits within the takt time requires completely different measures, and that's the job of managers, supervisors, and technicians

For instance, if a person requires 70 seconds to finish from process one to process five, while takt is 60 seconds, they would exceed takt by 10 seconds. There should be no need to explain that we must never omit those 10 seconds. The people should do their work normally, and the line should stop for 10 second intervals every time to make products with good quality.

It is the job of the supervisors and the technicians to implement improvements to make sure that any person can finish the processes within 60 seconds working at a normal pace. By cutting down unnecessary activities or reducing the distance they walk, with improvement, there would be no need to stop the line.

If you try to eliminate the line stop without improving the working process, this will naturally result in a lower quality of products; this must be strictly avoided.

In the case B, by reducing the number of people and inventory, we see that many defects were produced frequently. These were reworked within the production process rather than solving the root cause.

The following process fixes the defect caused by a preceding process without giving feedback. For example, there is a mismatch with the tapped hole due to a design problem. You fix it with the correct screw holes in your process.

Because the problem is managed, the root cause remains unsolved. The additional labor and inventory for these makeshift repairs increase the cost. When these problems are made obvious because of labor reduction process, it is a chance to improve.

Supervisors and technicians must return the defective products to their respective departments. They should visit the preceding process to pursue the root causes behind these defects, and solve them starting with understanding the root cause.

This can be compared to taking pain medication for chronic appendicitis, instead you decide to get surgery and fully recover.

This concept also applies to solving the defects caused by machines, equipment, molds, and jigs, which we have described earlier. If we find that the equipment is causing a defect, it is important that we stop the line immediately and to identify and eliminate the root cause.

If we end up doing in-process readjustments on our own just because the maintenance department wouldn't respond after we contact them, we may start to fix the problem within our own production process, and this imperceptibly becomes a part of the regular process.

We shouldn't stop with a single written request or a single phone call. We must keep taking measures patiently until we get a product of perfect quality.

Defects and Inspection

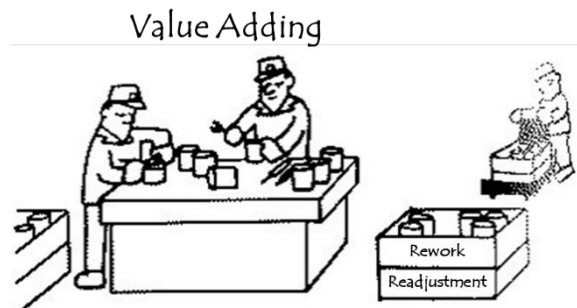
Let's think about why sending out a defective product is unacceptable. If the assembly line which is the final process of production, sends out defective products, then it is highly possible that a defective product will reach the customer.

This is unacceptable, because this affects the trustworthiness of the company and the credibility of the product, more than anything else. Normally, defects are discovered by inspection and reworked before they are sold to the customer. The more the company is determined not to put a defective product on the market, the stricter their inspection becomes, and thus the more frequent rework needed. This only increases your cost.

The extra inspections done is work that does not result in any value being added to the product. Manufacturing can build gauges into their jigs and fixtures as an automatic inspection process

that reduces the possibility defective products being used (this error proofing is called *poka-yoke*). This is done to ensure that every single product is of good quality.

We believe that the readjustment should be regarded as work that should not have existed from the beginning. The more people you need for inspection and rework outside of the process, the lower value-added ratio in the plant becomes, and the more the costs will increase.



You cannot say, "*This product is expensive because we test it 10 times*". This will not be accepted in the market. Work that does not add value is just an *unnecessary activity*. It is something that should be eliminated from the beginning.

Even if you omit the unnecessary activity within the direct work process and achieve some labor reduction, if you produce defective products, then inspection and rework increase the labor requirements. If you consider this from the point of view of cost reduction, the result may come out even, or even a loss; this result is far from the original purpose.

For this reason, we think in the following manner:

The inspections outside the process as well as rework are both unnecessary activity. We want to eliminate them as much as possible. Inspection should be the minimum necessary, and no rework required. You inevitably reduce labor.

It is important to be aware that large labor reductions are possible by eliminating inspection and rework operations, as you strive for improvements.

The in-process inspection should be considered as follows:

It is necessary for the people to check the quality of products they made, namely the 100% in-process inspection. We regard the subsequent production process as the customer, so not a single bad product should be released.

It's important to be creative with all the various aspects of inspection methods. In addition to visual inspection or gauge inspection, error-proofing must be considered.

As for lot production such as that using high-speed automated presses, we accumulate say 50 or 100 pieces over the chute, and inspect the first and last piece. If they are both good, they go ahead and get transferred to the pallets. If there is a defect with the last piece, examine from which piece the failure began to occur, get rid of them, and you must take measures to prevent such defects from happening in the future.

This is a type of total inspection. Even in a fast, quick production process, you should not think that you can conduct only sampling inspections. It is important to build in the quality within the production process.

The Goal of the Inspector

Let's approach the concept of "*inspections performed by an inspector.*" The job of an inspector is the determination of whether a product is defective. They sometimes think that is enough to just pass the results back to the previous step.

This is not enough. The inspector should be responsible for doing on-site analysis of why the failure occurred, and as much as possible, for determining the root cause, and stop the reoccurrence of this problem.

They should not merely be an inspector who performs a pass or fail test. Instead, inspector must be a teacher who explains what was wrong and teaches you so as not to repeat the same mistake.

For example, if a matter that came up was a mistake during parts assembly, the cause might not be just as simple as "*I was careless*".

There might have been many various causes, for example: the parts were not lined up in their assembly order, the line stop button, or the call button was too far away, or the work instruction information was difficult to read.

Without identifying these causes and taking the appropriate action, failures will not be reduced. The purpose of the inspector's work is not to toss out the defective product but to reduce them to zero. *Their work is evaluated on fulfilling this task.*

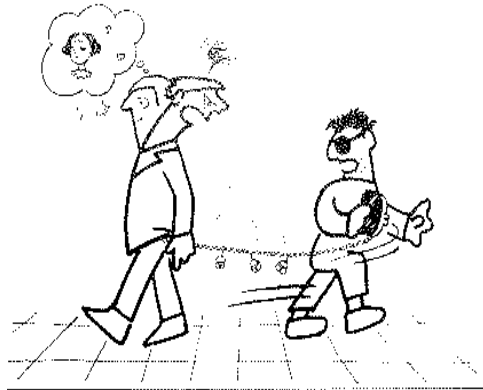
Error-proofing (Poka-Yoke)

During a production process, to build in high quality, we must think about which key points people need to check, which places do they have to measure, or when they need to replace cutting tools.

Devise ways of using jigs and fixtures so that products sent from the previous step are checked spontaneously. This means that you incorporate error-proofing (*poka-yoke*) within the production process to discover defects.

It is important to standardize error-proofing to make sure that even if people change, stable quality is produced.

Even the most careful people sometimes can make inadvertently mistakes, especially when they measure while performing other tasks, or doing the check by following a checklist.



That's why it's important to devise a mechanism that spontaneously eliminates defective products, operational errors, injuries or any other problems, even without a person paying attention to everything. The automatic discovery of any shortcomings, that is error-proofing.

To explain the mechanism of error-proofing (*poka-yoke*) a little more specifically,

- (A) A mechanism where products cannot be mounted in jigs when there is an operational error.
- (B) A mechanism where the machine will not start when there is a problem with the product.
- (C) A mechanism where the machine will not start when there is an operational error.
- (D) A mechanism to automatically correct operational errors before proceeding with production.
- (E) A mechanism to detect problems made in the preceding process.
- (F) If some work is left incomplete, the next step will not start.

Mechanisms other than these can also be considered.

The following can be thought of as error-proofing (*poka-yoke*) methods.

1. Signs... visualize the fault with lamps and color coding to make it easier to be discovered.
2. Jigs... devise a jig so that it won't accept different products, or it doesn't operate when there is a mounting mistake, etc.
3. Autonomation... to stop the machine when a problem happens during manufacturing. This is sometimes not counted as error-proofing (*poka-yoke*).

Error-proofing is an extremely important part of building in quality during the production processes. The aim is reducing problems to zero. To achieve this aim, it's important to think of

methods like those mentioned above, and to find the places that are easiest to grasp, as well as those with minimum loss, when establishing error-proofing devices.

Section 4 – Labor Reduction and Safety

Safety First

There's a saying, "*there is no use crying over spilt milk.*"

Various assets such as machines, even if damaged, can be restored by spending money, but human body is less likely to become fully recovered once injured. A fatal disaster, such as accident causing death, cannot be compensated with money.

Safety must be considered before everything else. There is no labor reduction where safety is not considered. We must consider safety as a foundation, and then think of all possible ways for labor reduction within that scope to reduce cost.

Sometimes we see cases where the improvement routine isn't allowed to proceed in the name of safety. At such times, we need to go back to the starting point and review the purpose of the work. There would be no development or progress, if you think, "*let sleeping dogs lie.*"

Relationship between Labor Reduction and Safety

You will see that the seemingly conflicting values of safety and labor reduction are consistent with each other, because the labor reduction means also promoting the elimination of unnecessary, uneven, and unreasonable activities.

In other words, most of the accidents in production plants occur because of behaviors which can be thought of as activities that are unnecessary, uneven, or unreasonable.

We believe that doing or being made to do things that are unnecessary or things that are hard to do, leads to unnecessary activity, uneven, and overburdening actions which can end up in injury.

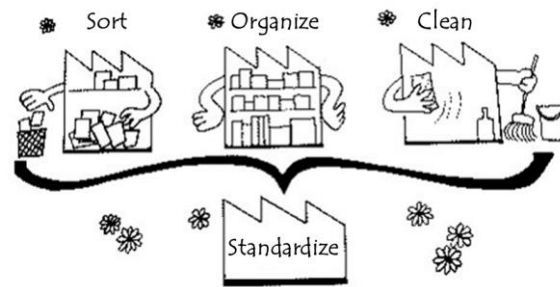
We could use the Japanese characters for the word "*Injury*" to write "*This is odd if I do say so myself*". *Odd* behavior that includes unnecessary activity, unevenness, and overburden is often the cause of injuries, and eliminating of unnecessary activity, unevenness, and unreasonable things leads to safety.

Workplaces where accidents are frequent, the day-to-day management is insufficient. These accidents occur frequently where the basic matters like 4-S, working procedures, and job instruction are not considered or even recognizable.

Workplaces that are passionate about their labor reduction and improvement routines, have fewer accidents. The more simplified the workplace is, the easier it is to manage. We emphasize 4-S as an aspect of safety management. 4-S is based on the people, materials, and equipment as they currently exist.

If the layout of the machinery and equipment is poorly designed; then practicing 4-S will not achieve its purpose.

There's a limit to how much you can sort something when it's moving towards complexity. It should not be overlooked that people, materials, and equipment on site are not independent from each other but rather intermingled in a complex manner.



For example, this is the vicious cycle: there are too many people → too many products (accumulate stock) → more people needed for clean ups, loading and unloading, storage or rework is required → even more people.

In this way, increasing one thing increases other things accordingly. In the workplace, complexity is in direct proportion. It is important in advancing the safety management to hold off this tendency for complication as much as possible by labor reduction.

We will briefly explain 4-S; it refers to sorting, organizing, cleaning, and standardizing.

- (A) Sorting (**Seiri**) - To divide things into needed and unneeded, and dispose of the unneeded things immediately.
- (B) Organizing (**Seiton**) - Arranging needed things in an easy-to-use manner.
- (C) Cleaning (**Seisou**) - To clean up and keep things tidy.
- (D) Standardize (**Seiketsu**) - To maintain the state of (A), (B), and (C).

Accidents in our Company

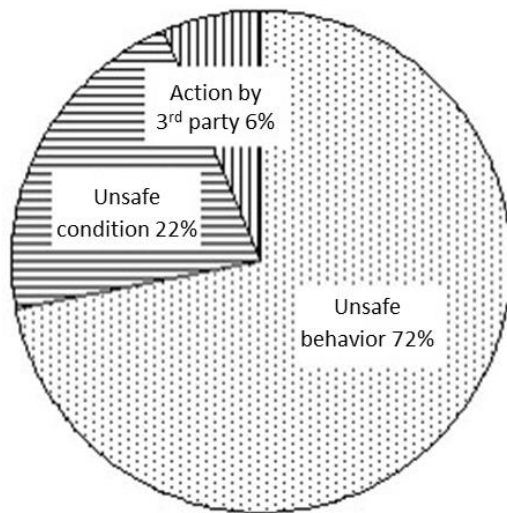
Looking at our company's accident status, we find that there were more than 800 occurrences of accidents per year. These accidents can be considered as just the tip of the iceberg.

There may be even more workplaces which are in similar conditions to ones where the accidents took place, as well as ones that are continuously exposed to danger even though accidents fortunately haven't occurred?

Two years ago (1970), 878 accidents took place. If we consider Heinrich's law⁷, there are probably an even larger number of people who experienced near-accident situations, even if it did not end in one. Such dangerous situations cannot be left unimproved.

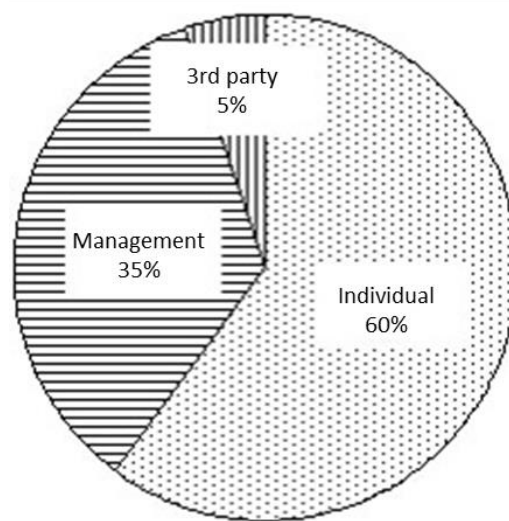
⁷ Heinrich's Law - In his 1931 book "*Industrial Accident Prevention, A Scientific Approach*", Herbert W Heinrich put forward the following concept that became known as Heinrich's Law: in a workplace, for every accident that causes a major injury, there are 29 accidents that cause minor injuries and 300 accidents that cause no injuries. (Also restated as 1-30-300 rule.)

Figure 2



Accident Cause

Figure 3



Accident Responsibility

Analysis of the main causes of accidents, as well as the risk conditions of the cause itself, we find that there were 142 incidents where safety wasn't taken into consideration; 77 incidents where maintenance adjustments weren't being made, and 14 incidents where there was no safety switch. These indicate that there is much room left for improvement.

If you look at the accident report of 1970, it shows many management responsible injuries linked to machine problems (20%); and the most frequent factor of accidents were forgetting to turn off the machine before working on a problem is (30%).

We have a problem in our current way of preventing recurrence of equipment issues, and in our accident preventive maintenance systems.

One problem behind all of this; equipment issues are all left for the production workplace to handle. It is necessary to address the problems not only within the workplace but also with management.

The pie charts in Figure 2 and Figure 3 show the number of accidents grouped by main factors and the responsible party. At 72%, "*unsafe behavior*" is the major factor for accident cause. Individual behavior is the dominant factor when we look at responsibility.

Within a simplified workflow, there are almost no unnecessary, uneven, or unreasonable movements, thus it becomes easier to perform. It is also easier to manage. In this sense, the more simplified the work becomes, the less risky behavior there will be.

This gives birth to the necessity of "*visual management*" for leaders. They need to be able to immediately recognize when people do unsafe, unstable actions. This is how simplification is connected to the safety aspect in labor reduction routines.

First step towards a safe production site

Safe work environment requires a company-wide deployment. The first step is to create a site that is free from the unnecessary, uneven, and unreasonable activity. We need to create an environment where it is easy to discover the unnecessary, uneven, and unreasonable activities.

(A) Under what conditions will work be carried out?

(B) What kind of procedures will be performed?

(C) What length of time for performing activity?

A job instruction manual that specifies these A, B, and C, should be defined (and maintained). Following effective job instructions will lead to creating a safe production site.

Our conclusion is promoting labor reduction will also lead to safety. To create a safe workplace, we should continue to expand our labor reduction activities.

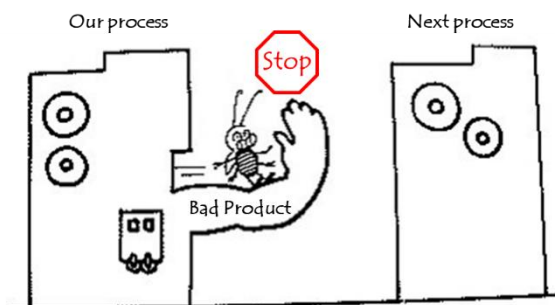
Following this, we would like to consider the issues of automation (*Jidoka*) and one-touch activation.

Simplistic Automation Causes Injury

Along with the increase of production in the past few years, the automation in our company has been rapidly developed. For the most part, the necessity of such automation has not been sufficiently questioned (i.e. there has been no thorough investigation regarding its unnecessary, unevenness, and overburden activities) thus, breakdowns occur often. To make it worse, there are high-speed machines without an automatic stop device, so it is extremely dangerous.

Because the automation was not designed as *Jidoka* (i.e. it was designed for labor-saving, rather than people saving), we still need to have someone to act as a guard, otherwise, it would come with the risk of the machine not functioning as required. “*Autonomation*” means that a machine is equipped with detection features and safety switches, which stop the machine when an abnormality occurs. This is important both from the safety perspective and from the labor reduction.

Automation – Automatic Stop Device



A machine that cannot automatically stop itself in the case of an abnormality, is at risk to bring about unexpected accidents. The more expensive the automated equipment is, the more we worry about it getting damaged, which ends up in our assigning people to guard it. The people

who guard it are rarely its designers, or its makers, and are not even the ones who do its maintenance, this also increases the risk of accidents.

For example, let's consider the following:

Near a completely automated conveyor, we often see a sign that says: *"Turn off the switch before climbing onto the conveyor."*

At first glance, it calls attention to the safety and looks as if safety is assured; but above the conveyor, there's a platform on which there are often bars for moving anything stuck in the conveyor.

A person may not want to climb on it, but if the need were to arise, they might turn off the switch before climbing. When the conveyor can't be fixed immediately (maintenance thinks it can wait, it is good enough if it just runs at 80%). And there is an expectation from management to ensure the production targets are met. Would the person still follow the rules and turn off the switch before climbing up on the conveyor?

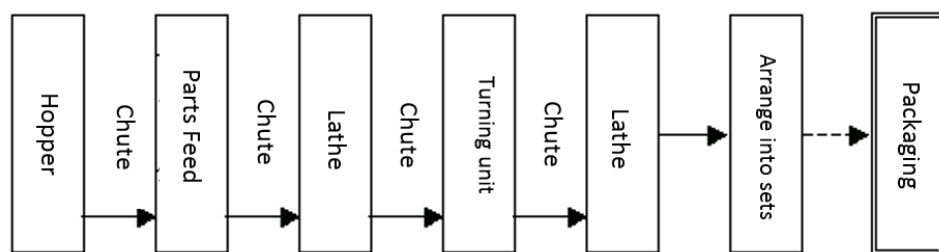
If an accident occurs, this will be reported that the person did not follow the rules. And the excuse might be given that the person did not receive enough training. But the real reason is the presence of incompletely automated equipment that it was left as it is.

For truly automated, self-activated equipment, must be structured such that running at 100% functionality is considered as a given, otherwise it must have the structure that stops automatically in abnormal situations.

In other words, the 20% of the conveyor stops that were recognized as acceptable, should be considered as the cause that lead to injuries.

Example from a machine factory

The following case took place in a machine factory. Cases like this are common problems in automated workplaces. Let's review the facts.



In the processes shown in the diagram, a person had their finger severed by a lathe.

Originally, the person's job was just to take the work that was automatically arranged into sets and pack it into boxes. Since there were about 10 lines like the line shown in the above

illustration, one person should have been enough to cover them. However, there were always three to four people on each line to improve the flow, otherwise scheduled production could not be met.

Because the flow through the chutes were irregular and the “No Work,” “Pass Confirmed,” and “Full-Work” switches were not properly maintained, when an abnormality took place, the machine wouldn’t stop.

The accident occurred when the person’s finger got stuck in a machine that was running “automatically.”

For this example, a person observing that workplace should question why one person could not do the job. Improving the flow on the chutes would achieve real labor reduction as well as creating a safer workplace; and more importantly, an accident could have been prevented.

Is ‘One-touch’ Activation a Risk?

We have explained that the concept of labor reduction and the direction we are headed, do not go against safety, but rather, they coincide with the development of the various conditions required to promote safety.

At the end of this section, we would like to mention an example of a stamping press where labor reduction also resulted in safety improvement.

One-touch activation for machines such as stamping presses has begun to be adopted. In the machine shop, it has been utilized for quite a while. This is because of the concept that multi-machine handling is a matter of fact, and has brought out many innovations which consider safety.

If we go with a conventional type of activation that uses a two-hand “hold-to-run” button until the press reaches the bottom dead center, that time used would be an unnecessary activity, and the extra walking time kills the benefits of multi-stage machine handling.

Why have the two-hand “hold-to-run” buttons been utilized up till now? They have been adopted to meet Labor Safety Regulations. According to the Labor Safety Regulations, *“Machines such as presses must include measures to prevent entrance of body parts in the danger zone when using slides or cutters. However, this doesn’t apply to presses structured so that the slides or cutters can be immediately stopped when a body part enters the danger zone.”*

The two-hand “hold-to-run” system merely complies with the latter part of this paragraph, but it does not fulfill the original purpose. Even if it has a two-hand “hold-to-run” button, if a third person is involved, then it does not work effectively unless the person who did the activation noticed the danger. The problem is not whether one-touch activation is bad, but is the machine

structured so that it can make an emergency stop when a body part is detected in the danger zone.

Progress in the development of devices such as activation buttons with safety structures, where the machine immediately stops if something approaches it, reduces the risk associated with activation using a one-touch system. The machine must be stopped in the event of a fault of the safety device.

Even better would be a machine that people can use without putting their body parts near any danger zone. Such concepts have been gradually getting introduced in stamping presses and automatic welding lines.

For example, there is a 250-ton press that is equipped with the copper wire at the lower bottom of the shutter. If something touches the wire, the limit switch will stop the press.

There is a new type of automatic welding machine with a platform around it. It will not operate while a person is standing on the platform.

Both are examples where a push button was converted into a one-touch button. These are still elementary devices, with further improvement, we will probably be able to make all activation buttons into one-touch.

Even if you stop thinking about the rationalization of the work, convinced that there is no choice for the sake of safety, you may notice that there are several options of a safe and rational way of working, just by going back to the fundamental purpose of the work.

Continuous effort put into coming up with working methods, and systems for equipment or devices that lie within the margins of safety, while choosing the ones with the minimum unnecessary activity, will surely become even more important in the future.

Section 5 – Labor Reduction and Human Relations

Respect for People as a Foundation of TPS

Labor reduction is often considered as speeding up of labor; but the Toyota labor reduction activities aim mainly the elimination of *unnecessary activity*, and it does not mean labor intensification or speeding up.

For example, a person must walk of five or six steps to pick a part for the assembly line, or they are going back and forth many times to the car. By redirecting those five or six steps, and the effort of going to and from the car, into work that clearly results in added value, we will be relatively reducing labor.

Like this example, everyone believes the work they are doing as part of the task, yet there are many unnecessary activities that do not add any profit. The idea is to eliminate these unnecessary activities (behaviors); shifting a person's energy to more useful work leads to respect for human dignity.

There is nothing more demoralizing for people who are offering their precious energy and time to the company, and being forced to waste it. Concentrating their efforts into work that's useful is the beginning of *respect for people*.

Human beings are motivated when they focus on meaningful work and are recognized for the value of their own work. On the contrary, when they are forced to meaningless work, consciousness of their work value becomes totally out of question. And of course, we can't expect them to be motivated (good morale).

There is a limit to the energy that can be produced by a human. To what extent have you been able to redirect this effort into useful work is linked to respect for people.

If a problem arises with labor reduction being labor intensification, or ignoring humanity, then there was something wrong with the way of reduction, or it is because of some misunderstanding.

Thinking from other people's point of view to build true relationships

When conducting labor reduction activities at the workplace, if they put themselves in the shoes of the people doing the work, then true human relations will be developed. There should be few problems like complaints of labor intensification surfacing.

If they stand in the workplace every morning to observe how they do their work and think from the people's point of view to achieve improvements, making a full commitment till the improvement will finally work, then the supervisor will not be isolated from the people in the workplace. We cannot build true human relations by mere superficial praise or flattery.

There is a story about Hirobumi Omatsu, a famous coach of the Japanese volleyball team. It is an old story, but I think it helps understanding the development of human relations in a company, so I quote the example below.

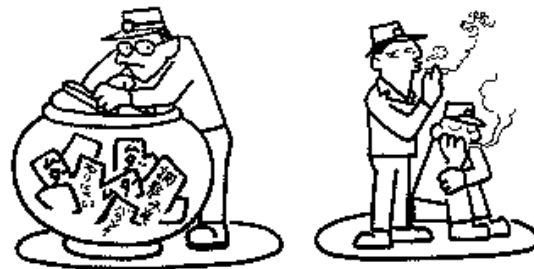
Coach Omatsu, never flattered or coaxed the team to encourage or motivate them. Instead, he thought from the player's viewpoint. By doing so, he established a relationship of mutual trust with the team. His rigorous training was based on mutual trust. In the game, he would make the best use of players and the players did their best to meet the expectation of the coach.

For a supervisor to have the workplace following their instructions, a relationship of mutual trust must be built between the supervisor and the operators. In other words, there must be true human relations.

To achieve this, the supervisor must have *respect for people* as a foundation, while constantly putting themselves in the people's shoes. They must train them while fully understanding them, stand in the lead, and proactively confront all problems, no matter how troublesome they might be.

What is especially important here is the fact that when there is any kind of problem in the workplace, someone is troubled by it.

For example, *"The work is hard," "This is dangerous," "It is difficult to adjust, producing too many defects despite our hard work," "The demand from the subsequent process is so variable that we can't decide on our priority order," "The Standard Work sheet is difficult to read and understand,"* and so on.



Everyone complains these kinds of issues at the beginning; however, when a prompt response is not taken, then people gradually cease to complain, and the problem left unsolved⁸. If you ignore these issues and just keep saying *"Reduce people!"* or *"Do improvements!"*, it only leads to growing discontent and complaints.

If a supervisor puts off the work they must do, and just pushes it onto the operators, it is only natural for the people, as human beings, to feel distrust towards them.

⁸ This is a source for many of 'people problems', like bad attitudes, especially towards managers. The leader has left the problem unresolved which makes for an unpleasant workplace.

A collaborative attitude of supervisors and staff to share the difficulty and hardship of people at the workplace, to think from their perspective, to solve problems together, will establish their trust in the supervisor. Which cultivates working together for improvements.

By following the concept of *“people must be treated as individuals, as human beings,”* which is mentioned in TWI’s “Job Relations (JR)” as a foundation, it is important to take care of various issues of human relationships to make full use of the people’s capability and abilities.

This spirit is the essence of human relationships, which has not changed and will remain unchanged. Aren’t we partially lacking in the senses of unity and trust, which arise from the experience of sharing the meals, struggles, and joys together. In the workplace, it is important to revive the sense of solidarity. The labor reduction activities must be deployed from that solidarity.

In doing so, it is important to let people participate in the activities so that they can share their opinions with us, and come up with ideas together. The concept of labor reduction itself is reasonable and convincing to everyone. If the reasons are misunderstood and are a hurdle for relations, it is often that the mutual trust has already faded. And the people don’t have the mindset to try to understand the aim of the supervisor.

As an example, let’s discuss a case at a factory when they improved the factory’s final assembly line.

(A) A brief explanation of the situation:

1. Late September 1946: the Takt (allowed cycle time) was decreased from 3.6 minutes per vehicle to 3.3 minutes per vehicle (without increasing the number of operators).
2. Through the end of September to late October: 120 - 150 minutes per shift of line stops occurred.
3. Many complaints from the operators were reported to the labor union such as: *“Too much work,” “This is labor intensification.”*
4. Mid-October: together with the technicians from the service department and we went to the production lines.
5. We applied improvements centered around the problems that people have difficulties with on the line.
6. The number of line stops during the mid-November have been reduced to 20–30 minutes per shift.
7. Approximately 100 improvement activities, large and small, were carried out. Only the first 20% of the improvement were carried out by the supervisor, the remaining 80% were applied by the people at the workplace.

(B) The reaction in the workplace:

1. The complaints, which were reported from early October to late September, are no longer reported at all.
2. Many ideas for improvement were proposed by the operators in the workplace, and are getting implemented one after another.
3. In the evaluation meetings of team leaders and supervisors held in November and December, the following opinions were given:
 - a. The people in the workplace became more motivated. They developed positive attitudes.
 - b. There is still some unnecessary activity that needs to be improved.
 - c. Team leaders need more time to spare so that they can sufficiently focus on improvement.
 - d. It was very good that their requests have been immediately responded. From now on, we'd like the technical service department to fix problems at the workplace immediately. They should not leave problems unsolved.
4. In just in one month, the motivation of the workplace has been raised a lot. Complaints about "*labor intensification*"⁹ completely disappeared.
5. Recently, this is how the workplace feels:
 - a. Whenever they spot the improvement staff, they come and discuss immediately whatever problems they are facing.
 - b. Suggestions for improvements are proposed by operators: "*It would be better if we do this.*"
 - c. In implementing the improvements at the workplace, the operators decide the arrangements on their own, and carry out it together after regular work.

⁹ Speeding up of work, making work harder.

Mutual Trust Starts with Participation in Improvement Activities

Supervisors and staff must respect the proposals and discussions coming from the workplace, and collaborate with operators to promote labor reduction activities.

All people involved can enhance their motivation towards improvement and have a sense of participation in the improvements. Reducing the feeling of dehumanization that comes from the monotony of the work.



From the operator's point of view, they realize that they can improve their workplace by themselves and feel the sense of achievement and satisfaction that they accomplished the improvement of their own. This will enhance their confidence, encouraging them to find further possibilities of improvements.

The sense of achievement, sense of satisfaction, and self-confidence are keys to overcoming the feeling of dehumanization. This is a source of motivation and eagerness that *"we can do it."* In a workplace with this kind of awareness, you can expect natural rise of morale.

An example: When the production of a machine factory needed to be increased, the front supervisor gradually increased the speed of the assembly line conveyor, while the rear supervisor consulted everyone about what to do.

In this example, the front supervisor made remarkable increase of production in the beginning, but were stuck after a certain point. The rear supervisor that took some time before implementing the improvements achieved a final faster production rate.

Supervisors and staff should always try to promote the improvement activities together with the people who work there, and spare no effort to create an environment and atmosphere in which everyone can participate in the activities.

Signals are used in every plant to create the conditions that enable visual management. They make it easier for everyone working at the place to find the problems and improvement points will be easy to discover. To enhance the desire of everyone to participate, we actively work to include all the people in labor reduction activities.

In our plants, with the supervisor at the center of the workplace, people propose constructive ideas to each other, such as: *"This process will be better done by this way"* or *"There will be less unnecessary activity by doing this process in this way rather than that."*

This eagerness to pursue their aim to reach satisfaction and fulfillment, their morale is the evidence of the smooth human relationships in the workplace. It is an embodiment and manifestation of our founding spirit.

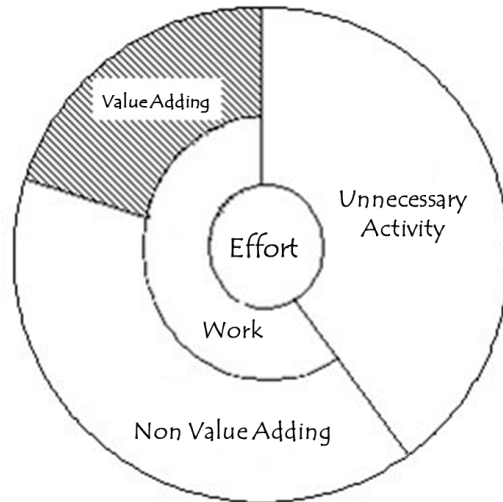
Section 6 – How to Promote Labor Reduction Activities

Awareness of the details of the work in the actual workplace

To promote labor reduction activities, developing awareness in the workplace is an important first step.

For example, if you think, *“The line utilization rate is fairly high under the current operation system. The manufacturing failure rate is within certain range. All in all, it looks like working quite good,”* then, there will be no motivation for improvement. You are nipping off the buds of improvements.

No matter what kind of workplace it is, when observed closely, what is said to be “work” (or effort) can be divided as shown in this diagram.



The types of work are:

A. Unnecessary activity – It is not necessary in any way when doing work. They can be eliminated immediately.

Examples: Waiting time, meaningless transportation (piling up of partially-finished products, the labor of transporting two or more times, or changing hands.)

B. Work that produces no added value - Work with no value-added is something that must be done under the current working conditions. (It's better to consider this to be unnecessary activity as well.) To eliminate this, we must partially change the conditions of the workplace.

Examples: Walking to pick up parts, unwrapping the outsourced parts, taking out parts from a large pallet in small amounts, or operating hold-to-run buttons.

C. Useful work that adds value.

Apart from regular work, things that are done as exceptional operations outside of the standard operating procedures, like fixing a small problem that happened to the equipment or jigs, or readjusting defective products and so on reduce the value adding ratio.

The proportion of activities that directly increase the added value is surprisingly low¹⁰. (All elements, except for value adding work, only increase the cost).

Labor reduction means increasing the ratio of the *value-added work* (profitable work). Our ideal goal is to get as close as possible to 100% *value-added work*.

Unnecessary Activity of Over-production

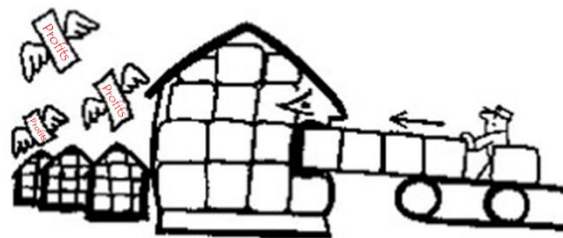
In the previous section, we discussed that if we closely observed the work in the actual workplace, we will find that it includes many other activities than *Value Adding* work. If these unprofitable activities are clearly visible to everyone, it will not be difficult to study measures to improve the situation.

For instance, within the work done by a group of people, everyone spends some time waiting, or doing meaningless temporary movement of items. In a production line, if we eliminate this unnecessary activity and redistribute the work activity, reducing the number of people won't be that difficult.

However, this is rarely the case. It is not that there is no unnecessary activity; rather, it is that the unnecessary activity is hidden and difficult to see. The most common issue is the excessive advancement of work. Because operators work on the next unit during the time that should be counted as waiting time, the unnecessary activity of waiting becomes invisible. This process repeated over and over again ending up with stocks piled up in the middle and end of the lines.

To make matters worse, additional tasks arising from the over-production, such as moving these stocks to other places or re-stacking them neatly make it even more difficult to distinguish the other unnecessary activities.

In TPS, this is called unnecessary activity of overproduction, and among the numerous types of unnecessary activity, we regard this one to be the one that must be avoided the most.



The classification and explanation of the types of unnecessary activity are described in detail in Chapter 2, so we will omit it here; however, the unnecessary activity of overproduction is different from the other types of unnecessary activity, in the sense it hides other types of unnecessary activities.

¹⁰ There are claims that Toyota calculates that less than 90% of their activities are value adding.

From the viewpoint of improvement, the other types of unnecessary activities give us clues, while the unnecessary activity of overproduction conceals them, and works in the direction of obstructing improvement.

The first step to promote labor reduction activities is to eliminate the unnecessary activity of overproduction. For this purpose, the line should be improved by establishing standards that prevent overproduction and by imposing restrictions with the equipment (such as a full work equipment). Only when you accomplish this, will the flow of things return to its original form.

As things that you need began to be produced individually, as they are needed, unnecessary waiting time becomes visible and noticeable. Once the production line reaches such a state, it will be easy to apply the routines mentioned at the beginning of this subsection: eliminate unnecessary activity → redistribute work operations → decrease the number of people.

The Concept of Takt¹¹

To avoid overproduction and produce only what is needed, when it is needed, it is necessary to know the time “*when*” it is needed.

The concept of “*takt*” is important. Takt is the amount of time that should be allotted to produce one product. For instance, the product must be made in X minutes and Y seconds. This is derived from the required number of products.

In a simple formula, it would look like this:

$$\text{Takt Time} = \frac{\text{Operating time available per day}}{\text{Required number of products per day}}$$

A common error in calculating takt happens when you use current equipment capacity, facility capacity or available labor.

For instance, “*Our equipment has this much capacity, and we have this many people, so how many minutes would it take us to produce a single item?*” This is what they consider their Takt time to be. This is a completely wrong understanding of the concept of takt in TPS.

There is less flexibility in the equipment capacity as they are fixed to some extent, but line capacity may be increased or decreased by adjusting the number of people. Work can be carried out by the minimum number of operators only when the takt is calculated based on the correct understanding of it.

¹¹ Takt is a German musical term for beat. Here we use it to pace the required production rate.

If you try to produce as many products as possible at the maximum capacity under ideal conditions, you will end up with overstock and this will not lead to any labor reductions.

When you have determined the appropriate takt, what should you do so that it's accurately followed? Details will be explained in another section. Follow these guidelines:

- (A) In the case of line work: draw a separator lines on the conveyor to mark a section designated to one product. The conveyor should flow keeping to the standards: one product in one section.
- (B) In between the processes: employ visual signaling methods. For instance, place a sign or tag (*Kanban*) and make it a standard not to be allowed to conduct the work before the tag (*Kanban*) is taken off.

For more details on *Kanban*, see Chapter 4.

The role of *Kanban*¹² is in the process to prevent overproduction by informing the required needs to the preceding process, this should be well understood.

Redistribution of Work

By limiting the ability to overproduce, the unnecessary activity of waiting will become visible in the workplace.

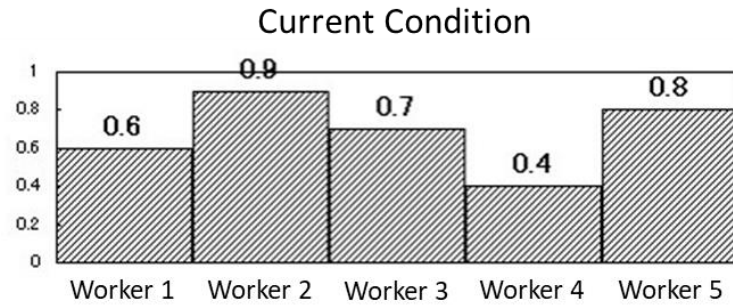
Next, work that does not directly produce added value as defined in Section 1 (requires awareness of the details of the work done in the actual workplace) should be improved immediately, starting with those that don't cost much money and don't affect the preceding production process.

For example, when people need to walk to pick up parts, we should move the shelves that contain the parts and eliminate the walking time. After adjusting the line like this, we redistribute the work of all the people.

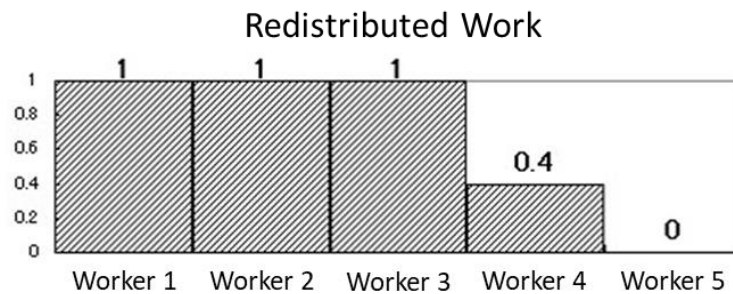
When distributing work within the takt time, only assign value adding work and the work that can't be currently eliminated.

If we simply explain this using a diagram:

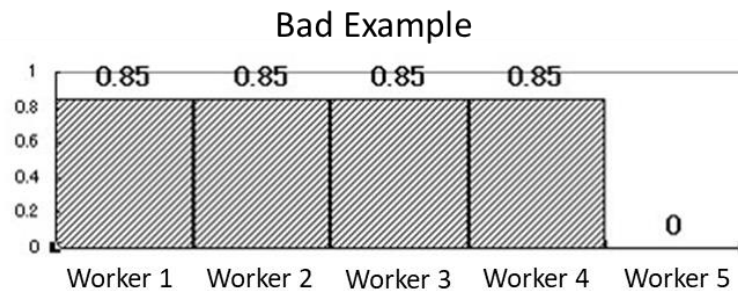
¹² The kanban card is a demand signal to the preceding process that materials have been consumed and need to be replenished. The Kanban System was developed to control inventory visibly as would happen on a dedicated conveyor system. With a conveyor, you can visually see when you need to add materials to the feed and it limits how much you can add. Kanbans should not be used between activities that have visual contact.



The current work distribution becomes;



It is important to note that *the surplus time of worker 4 must not be equally redistributed.*



The reason for this is that the equal distribution of the surplus obscures the unnecessary activity that had been clarified and made visible to everyone where the problems were and conceals the need for further improvement.

Furthermore, if all four of the people continue working with the surplus of 0.15 (day), which is the unnecessary activity of waiting, even if it is only for 10 days, this will reset their pace of work. When the next improvement is carried out, they may have psychological resistance, feeling that their work has been increased and become harder.

In our example, as the result of putting together and redistributing the work that had been done by five people, the same work is now carried out by 3.4 people according to calculations. Since there is no 0.4 person, it is still one person.

In other words, it's a situation where we've been able to reduce one person, and reduce the work of another to only 40%.

As described earlier, by reallocation of the work, one person has been reduced. The next thing is to reduce the odd work of person number four.

Our next goal is performing improvements that eliminates the 0.4 person's work. What shall we do to fulfil this 0.4 work without using a person? We focus on that and come up with a variety of proposals.

Among these plans, there are ones that cost money, like "*autonomation*" of equipment, as well as plans that don't cost much money like making a small chute to shorten the waking time, or making the pallets smaller, and putting parts within reach.

At this stage, it's important not to choose an excessive plan. The purpose is to elimination of 0.4 person's work. The most appropriate plan is the one which cost least and is easiest to implement.

With this, the work that used to be done by five people, using a relatively small sum of money, can now be done with three people.

Next, re-examine the production line one more time. If you look closely, you may discover other unnecessary activity that has been overlooked. Or possibly, you will find work which is considered non-value-added, which we have no choice but do in the current situation; but since it doesn't add value, it makes you rethink about whether it's possible to stop doing it somehow.

Once you gather these issues together and improve the situation, the challenge is, "*can we take out another person?*" This time, it won't be as simple as before.

If all the possible plans cost too much or affect the preceding/subsequent production processes, perhaps it cannot be realized now. By persistently observing the workplace every day while being aware that "*there's a problem here,*" you might come up with a great idea.

It is also possible that the awareness of the issues would help achieving the improvement as new needs arise in the future, for example, when the takt has been changed according to the sales or when there is a facility change due to a model change. It is important to work patiently without giving up, even if it cannot be improved immediately.

Sequence of Improvement

In the previous section, we explained the method to proceed with improvement in this order: Redistributing work by eliminating unnecessary activity → Applying improvement to labor fractions → Further review.

Looking at this as described in the Section 1, the process can be explained:

- (A) Immediately eliminating unnecessary activity,
- (B) Reduction of work that does not add value, starting from the ones that can be most easily handled,
- (C) Leave the Value Adding (VA) work as it is.

When proceeding with labor reduction, difficult parts in (B) can be handled by spending money. In the case of the VA work in (C), it is possible to cut down on personnel by automation (Jidoka) if the need arises. In other words, all the works (A), (B), and even (C) can be subject to improvement. Depending on the situation, we might need to implement all of them.

When performing labor reduction activities in the workplace, it does not necessarily have to be done in the order as explained in the preceding paragraph, as far as you keep the following things in mind:

It is more likely that you can improve faster when you conduct the activities concurrently. What you should be careful about is the order you carry out: you should conduct the work improvement first, and then do the equipment (or facility) improvement. The improvement plan can be roughly divided into "*work improvement*" and "*equipment improvement*."

"Work improvement" includes setting the rules of workplace, redistributing work, and specifying the locations of things.

"Equipment improvement" includes installing devices, and automating equipment.

When carrying out the improvements, keep in mind that you should start with the work improvement and complete all work improvements, before carrying out any equipment improvement.

The reason is:

- (A) Equipment improvement costs money. The objective is to have fewer people. If we spend a large sum of money on equipment improvement, when we could have done fine with work improvement, it could be said that we chose the wrong method.
- (B) Equipment improvement cannot be undone. Equipment improvement can end in failure even if we thought it was the best way during planning.

Essentially, there are some elements involved that can be carried out only through trial and error. If something goes wrong in work improvement, you can easily make changes in the specific part of the process. On the other hand, however, if anything does not work well in the equipment improvement, you will then lose the money that has been invested for the improvement.

(C) Equipment improvement in a workplace that didn't fully undergo work improvement, is likely to fail.

Equipment improvement can most likely fail due to its inflexibility, when implemented in the workplace where organization and standardization of the working processes has not been fully done

For example, if we automate a press in a workplace with poor raw material management, the molds and automated equipment could fail right away because of contamination by foreign materials. If we end up assigning someone to watch over the automated machine to solve this problem, then this can't be considered as labor reduction.

For these reasons, TPS emphasizes you do the work improvement before the equipment improvement. It should be noted that this concept is also true in the case of promoting "*automation*" (Jidoka).

Originally, autonomation, or automation with a human touch (Jidoka), was equipment improvement with an objective of reducing the cost (labor reduction). However, there are a relatively large number of cases where autonomation became something like an objective, and implemented without considering the progress of work improvement in the workplace.

If the work improvement was insufficient, the expensive autonomation causes troubles and unnecessary activity; for instance, making defective product by lots, considerable decrease of the operation rate due to frequent machine troubles, or, the machine cannot be operated without an operator. Careful consideration and reflection are needed on how to proceed the autonomation.

Other Important Points

We have explained above about proceeding with labor reduction. In the end of this section, we want to add a few more points. Part of this will be discussed further in the next section, we will summarize only the main points.

Make the Problems Visible

It is important to keep the workplace well organized and standardized so that the problems are clearly visible and recognizable to everyone. If the problem is clear, it makes it easy for everyone to think of improvement ideas together.

Do not be Afraid of Stopping the Line

In the process of improvement, a temporarily confusion may cause line stops. If you are too afraid of stopping the line, it will not be possible to improve the current situation. Instead of fearing line stops, you should consider them as the springboard for further improvement. It is important to take a quick measure so that the line can be operated without causing stops.

During the process of improvement, it is necessary to not hesitate to stop the line to clarify and solve the problem. This may cause a temporary decrease in the production quantity and the staff may feel upset.

In the assembly line of a certain company, a supervisor achieved improvement by following this method. During the improvement process, the production quantity that had been 2,000 pieces per day went down to 1,500 pieces per day and stock was quickly reduced.

Having been criticized by their boss, they started to waver inside. They kept believing that this was the only way to achieve improvement. Because of this persistence, in about ten days, that line became able to assemble 2,600 pieces per day.

While the supervisor of the neighboring line worried about how stopping the line even temporarily will decrease the efficiency, and will be a loss to the company, so they didn't allow their operators to stop it.

The improvement of the former line made the unnecessary activity of the latter line obvious. The difference between the lines became large enough for everyone to notice.

This is an example that has happened, but you must think thoroughly about how, inside Toyota, or even the lines of its suppliers, the lines that have relatively little unnecessary activity and high efficiency, don't fear line stops.

Stopping the Line



How to Reduce Labor Required

Even you have a great improvement plan, it is difficult to realize it without the cooperation of the operators. To make the significance of the activities fully understood by the people and to obtain their cooperation, the following points must be considered.

- (A)** Help the operator recognize that they have time to spare. An operator who has waiting time, should be allowed to idle for a while without doing anything. By doing so, it will become apparent that they have some spare time. There will be no strong resistance or opposition when they are assigned an additional task.

(B) When reducing people, reduce the best personnel first.

We often make the mistake to reduce the person who is difficult to control or those who are not good at or not familiar with the operations. By doing this, the person concerned may never show any development no matter how much time passes, and will become conscious of the opinions or expectations of others, and will resist them.

Reduction of the person who is doing poorly leads to a decline in morale. By reducing those with the good results, you can obtain active cooperation of others.¹³

¹³ Takehiko Harada details this in *“Management Lessons from Taiichi Ohno”*. The best performer is removed and assigned the task of assisting with further improvements; this becomes a reward for making improvements.

Section 7 – Labor Reduction and Supervisors

Role of the Supervisor

It is said that "*reducing the cost 10% is equivalent of doubling sales.*" The cost reduction activities at the manufacturing sites are essential, if neglected, it can shake the company's very foundation.

The role of the supervisor, who is responsible for the actual practice of promoting the activities, is very important and the weight of their contribution to a company is great.

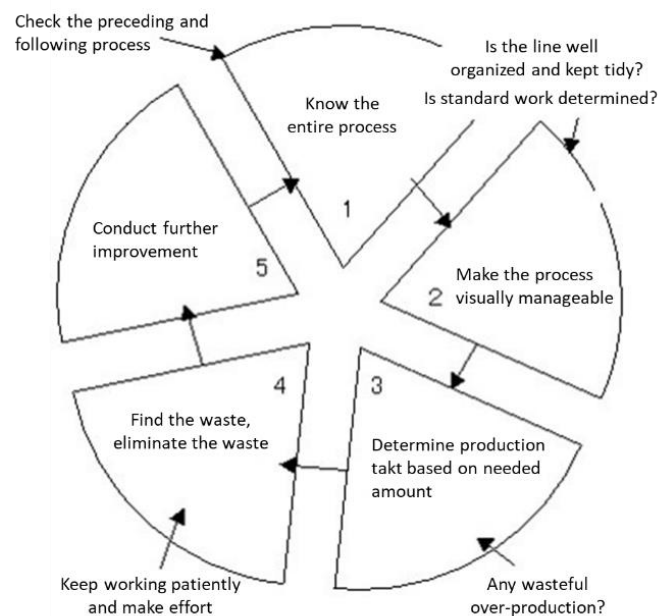
Recognizing this, we are promoting TPS in Toyota's factories, and executing using the following method of analysis.

When following this circle, we would like to discuss what kind of resolve the supervisor should have when going through this cycle, and what they should do to make sure that Toyota's labor reduction routine is carried out, in a manner that complies with the company's goals.

There are two fundamental roles of the supervisor. The first is quality assurance¹⁴ (see Section 2 on Labor Reduction and Quality), and the other is to conduct continuous improvement activities for labor reduction.

This is not as simple as you might think, because it contains elements that seem contradictory to each other at first glance. While you need to secure the quantity and quality, at the same time you need to achieve this with the minimum number of people and equipment.

By putting too much stress on securing the production quantity, they would most likely start thinking that they must avoid line stops at all costs from the beginning. So, the people, equipment, and stock would increase. Ultimately the cost will go up, as they have strayed from the original objective of the company's routine.



¹⁴ This is more than identifying defects or problems, it involves problem solving and implementation of containment and corrective actions.

As hard as it may be, the supervisor is required to make effort to improve their production in a direction that satisfies both.

Management of Abnormalities

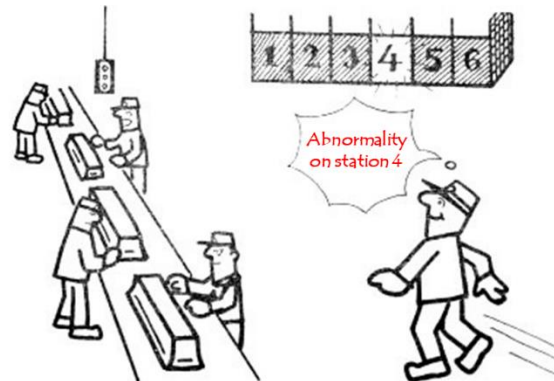
What methods should a supervisor use to apply improvement to the line so that goods that satisfy each of the quality, amount, and cost, are produced?

First, let's think about the manner of management. Even though we only thought of few, there are many things that are subject to the supervisor's management. People, work distribution, teaching the "*know-how*" and quality, accomplishing or adjusting of the production plan, equipment, safety, inventory, arranging raw materials, setup, etc. It would be impossible to fulfill these at once by yourself.

In TPS, it is taught to standardize these elements, then focus on the exceptions that deviate from the standard. This is called the management of the abnormalities.

Standardization in terms of the operation phase is deciding on standard operating procedures and to follow them. In terms of material and stock management, it is to clarify their designated location and quantity. In the aspect of production instructions, it refers to "*kanban*," while, for safety, it's about criteria of handling. In short, standardization is to set guidelines for each phase of production process to determine how each task¹⁵ must be done by each operator.

By organizing every aspect of the manufacturing, setting the standards, and making sure people follow the standards, you can spot those which deviate from the standards as '*problems*' and focus on solving them. The more thoroughly you apply this method, the more it becomes clear what the supervisor must do.



The first thing you should do as a supervisor is to organize the line you are responsible for. To put this in a more detailed manner, this would mean establishing standards, deciding on where materials and parts are to be left, deciding on the production quantity, setting up *kanban* (stock control signals), and installing call buttons, line stop buttons, and *andon*.

Once the standards are set that show the supervisor's intention of "*how to do the job correctly*", then the supervisor must observe what has happened in the workplace as a result.

¹⁵ This is the Job Instruction portion of Standard Work. Even the training method is standardized.

Then, supervisor needs to take necessary measures to correct the abnormalities that were not originally intended. It is important to distinguish what kind of activity is normal and what is abnormal.

Although the supervisor may have clearly defined the standards, if they cannot recognize what is abnormal or neglect to correct these abnormalities, then they are not suitable to be a supervisor.

It is important to devise ways so that anyone can tell that there's an abnormality right away. Being able to see the abnormalities is the first step towards improvement. Let me explain this with some examples.

If everyone does their work according to standards, and there are zero defects, they are supposed to be able to produce the line's daily required quantity of products by the end of their working hours.

Since the distribution of work is not perfect, it is unlikely that well-balanced workload has been assigned to all the people from the beginning.

For example, operator A has a slight amount of surplus time. They are waiting every now and then. Or stocks are often piled up behind them. Or they begin to do tasks that are not included in their standard work. These are all abnormalities; operator A *doesn't have enough work*.

In another case, operator B doesn't finish their work within time. They stop the line, or they cut corners and produce items with defects. Those are also abnormalities. Operator B has *too much work*.

If operator A can do operator B's job within time, then that means that the supervisor's method of teaching was unsatisfactory.

Operator C is helping a machine with an automatic feed. (This is an action outside of the standard operating procedures.) This is an abnormality. When supervisor examines the abnormality, they find out that the machinery is wearing out (or out of adjustment). It produces defective products unless the operator assists. The supervisor should immediately contact the technicians or the maintenance personnel and have it fixed.

Behind the line, there are products that have no *kanban* (work order sign) on them. This, too, is an abnormality. Probably a wrong pace was set to produce too quickly, or maybe all the people have too much time to spare, or they didn't stop producing items although the following process stopped receiving it as something wrong had happened (violation of the kanban standards). In any case, none of these should be ignored.

These are just a few examples. All of them should be considered by the supervisors who establish the standards, to be abnormalities. In most cases, if the root cause of the abnormality is investigated, it would often turn out that the standard couldn't be kept because the standard

itself was overburdening, the parts or the raw materials were defective, or the equipment was malfunctioning.

The occurrence of line stops, and defects are directly related to quantity and quality, so we can immediately tell that they are abnormalities.

Abnormalities which would increase the cost are overlooked or thought of as a secondary matter because they are small unnecessary activity or violations of the standards. These are all important clues about improvement that lead to cost reduction. No matter how little they may seem, we need to avoid overlooking them.

Stopping the Line

The first step of improvement is to learn to see the abnormality. If you find the abnormality and clearly understand what is the problem, 50% of the problem is solved.

"This part is wrong. This has to be fixed." Once the problem is identified, various ideas how to fix the problem will be proposed by experts who have unique knowledge and skills. By gathering the collective knowledge of the operators, many more methods will come out.

You then choose the least expensive, most effective solution and carry it out. *"Making the abnormalities visible and noticeable"* is an essential principle for proceeding with improvements, and is a very important matter that the supervisors must devote all their energies.

Next, we want to talk about the supervisor's attitude necessary for this. It is important that you should always be ready to stop the line anytime. This may be a difficult thing for the supervisor to do; but it is necessary.

When you don't hesitate to stop the line, you can recognize the problem with the production process and the poor quality of the preceding process. Also, by telling your operators *"if you can't do it, then stop the line,"* you can encourage them to carry out the improvement activities more actively.

This may sound like a paradox; but the aim of stopping the lines is to improve them (stronger and better) into lines that don't need to be stopped. The purpose is eventually to build up an ideal line.

This is for this purpose that you stop the line with financial loss at stake. Therefore, when the line has been stopped, supervisor must do their best to solve the problem fundamentally (root cause).

A supervisor who can't say *"stop the line,"* as well as a supervisor whose line was stopped 2 or 3 times for the same reason both get an **"F."** Instead, to solve the problems those have become

clear by stopping the line one after another and eventually improve the line as productive as unstoppable is the important attitude and mind set as a supervisor.

It must be kept in mind that *“a line which does not stop is either perfectly good or extremely bad.”*

Implementing the Improvement

Next, we would like to talk about implementing improvements. What we’ve talked about so far is discovering problems. To improve the problem, there should be a variety of methods that can be considered.

In selecting the improvement plan¹⁶, it is significant to begin with improvement of work operations before starting the equipment improvement, as has been explained in Section 6.

When an improvement plan has been decided, and it is time to implement it, in many cases the results can’t be known unless you try and see for yourself.¹⁷

For example, you eliminate the unnecessary work and reduce one person from work that used to be done by four people, then redistribute the work on the remaining three, but there’s an extra 0.1 person’s work remaining. Things like that are common.

If you push it onto them, saying *“you should be able to do this,”* you will provoke the opposition of the people; they will consider this to be labor intensification (unreasonable – overburden). You have no choice but to focus on how to eliminate this 0.1 person’s work and to beat your brains out to find a solution. If you cannot resolve the issue on your own, you should consult your boss and your team. It is also useful to consult the technical staff.

In the improvement process, results are important. Once you set about the activity, work patiently until you finally achieve the reduction of a full person.

If you were going to save about 15 seconds of time, there are many things that can be considered like shortening the walking distances, having the parts storage location closer, making the pallets smaller, or possibly, changing the buttons to one-touch buttons (provided that you take measures to ensure safety at the same time), retrieving the products automatically, or hanging the tools from above; in short, the point is that you should narrow your focus, and continue to persistently think about what to do. In such a case, it is often that a tiny hint triggers a good idea.

¹⁶ Sequence of improvements should follow: People (job standards training), Process (layout, line balance, simplification, parts placement, jigs and fixtures), Machine (adding mechanized improvements – automating), Design (changing the design of the product).

¹⁷ If possible, design small experiments to test the ideas before implementation.

Another important thing is that the improvement plan must be well established, once incorporated, the improvement must become standardized as a new work routine. In this sense, an improvement plan is useless if it is a haphazard plan that cannot be stabilized.

To stabilize the improvement, when improving the equipment, jigs and tools, or chutes, for instance, you should follow up until they are fully used in practice. In the improvement of setups, such as exchanging the cutting tool or arrangement of molds, just setting a new standard is not enough; it is necessary to practice it thoroughly, including some minor revisions, until all the operators fully master the new work standard.

When an improvement activity is completed, and the work standard is revised, it is time to start another improvement. We must return to the actual workplace once again, and search for abnormalities.

The role of the supervisor is to turn the circle of: Standardization – finding abnormality – investigation of the cause – improvement – standardization. By continuously turning this circle, it is possible to fulfill the seemingly contradictory function, i.e., ensuring the quality and quantity and at the same time achieving the cost reduction.

Things to keep in mind as a Supervisor

Here, we will briefly talk about the conditions essential for the supervisor to fulfil their role.

The first one is observing the actual workplace constantly. A supervisor of the workplace who does not look at the line or who is indifferent to what is happening is not qualified to be a supervisor.

If they do not check the standards they determined, or cannot distinguish normal from abnormal, there is absolutely no possibility of improvement.

Second, a supervisor must control their team. This means supervisor must train the team so that they can work to the standard. As mentioned in the section on human relations, a good human relationship is not built by just earning the favor by being modest and reserved.

The supervisor who teaches and trains people who will eventually become their equals as supervisors one day, and the one who creates a strong workplace as a result, is one who can be trusted as a leader from the viewpoint of their people.

In TPS, we ask this question: "*Did you have to produce this inventory, or was it just produced impromptu?*" No matter how much material or manpower is available, it is important that you do not produce if it is unnecessary.

Supervisor should have the leadership to control the team as they need, stopping the team working or giving them the green light. Effective improvement cannot be carried out without leadership.

Third; the supervisor should look at all things from a broad perspective to make a comprehensive decision.

No matter how much you say that it was an improvement to your own production process, if it negatively affects the preceding or subsequent processes, or if you must switch from carrying out a troublesome production process to outsourcing, from an overall view, this can never be judged to be improvement.

It's very important for every line supervisor to take into consideration when they are managing their own line, and constantly make overall judgements by looking at all things from a wide perspective.

The best supervisor is one who can do the standardization and promote continuous improvement so that the line can work at its best performance even without them. The ideal supervisor can say: *"Please remove myself from the line."*

Supervisors Working on the Line.

In the end of this section, we would like to discuss the concept of a supervisor doing work on the production line.

We are asked: *"Is it right or wrong to let the supervisor to participate in the production line work?"*

In TPS, the answer would be: *"It is not good if supervisor is participating in the line work **all** the time; but at the same time, it is not that they do not need to do so at all."*

A supervisor that works on the line every single day is not different than an operator. Such a supervisor cannot fulfill their main role of conducting management, improvement activities, or staff training.

On the other hand, a considerable surplus of people is needed to make it completely unnecessary for a supervisor to work on the production line. Where a continuous effort is made for improvement, it is only natural that there are occasions in which supervisor must participate in the production line work. Do not participate in the production line reluctantly because you must do so.

As described in the first paragraph of this section, one of the important roles of the supervisor is improvement activity for labor reductions.

The reason why supervisor must be away from the production line work is to see the overall situation, the ultimate purpose of which is to carry out the improvement activities. At the same time, however, to perform the improvements, the supervisor must have full knowledge of the difficulty and procedures of the work.

For example, when new employees join the team, or the work process has been improved and the process has been revised, it is the supervisor who must show and train the best way with least unnecessary movements.

In addition, there are unnecessary, uneven, unreasonable tasks that cannot be recognized if supervisor is just looking at the line and does not join the practical working process.

Whether it's about guiding their people like this, changing procedures as they see fit, or searching for unnecessary activity, unevenness, or overburden in deep detail and applying improvement to them, there are times where they must work on the production line to figure things out.

In this sense, when you have no choice but to work on the line because an operator is absent, it is a good opportunity. It makes a great difference whether the supervisor joins the production line work reluctantly, or willingly takes the chance to polish their skills and to find possible improvements. This would not only promote the improvement, but also maintain production in the event of the absence of a skilled operator.

Although you may think it is a stern way of thinking, when the chance occurs to join the line, supervisor must always participate in the production line work, taking the advantage of learning, with a positive awareness of its significance for further improvement.

A Supervisor Must Fill All Roles



End of Chapter 1.

1973 TPS Manual

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トヨタ自動車工業株式会社

Chapter 2 – Efficiency

Section 1 – What is Efficiency?

As already mentioned in *Chapter 1*, it is most important for those working in manufacturing companies to reduce unnecessary activities and develop efficient production methods, which consequently reduce costs.

For that purpose, we must do our best to eliminate unnecessary activity daily and do our utmost to raise the efficiency of production through continuous improvement.

We generally use “*efficiency*” as a “*scale*” to measure how much our productivity went up compared to the past because of our efforts.

Regarding viewpoints and ways of thinking about various things related to efficiency, it must be well organized from day to day. If you misuse the way of using the standards, you cannot evaluate it correctly.

Not to mention, we might end up facing a situation where cost goes up despite the increase of efficiency.

In this chapter, I will explain the explanation focusing on the way of thinking about efficiency.

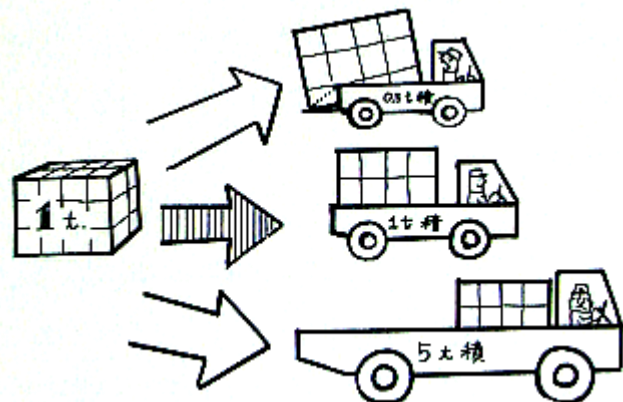
Effectiveness and Efficiency

I want to think about effectiveness first.

effectiveness means "the ratio of the work that the machine can actually do to the energy that it has supplied to that machine, so it is indicated by a number within 100%".

By applying this way of thinking to production, we refer to the effectiveness of production, which is “*for a certain product, it’s the percent ratio between the labor (energy) that turned into a product (or possibly the labor needed to proceed with the production processes), and the labor that was actually consumed to produce it.*”

To say that effectiveness of production is 50%, only 50% of the workers' efforts are producing the product (or to advance the process), and the remaining 50% is unnecessary. It means that it was done. Speaking of 80% production effectiveness, 80% of the workers' efforts are useful, so production effectiveness is quite high when compared to the previous example.



Effective production means that most of the effort that everyone makes is an effort that creates something (or increases the value-added), and this is positive.

When you consider it, everyone works as hard as they can when they start at a company. But when you look at the nature of their work, for example, piling things up here, lining them up there, carrying parts off the pallets in small amounts, arranging them together; there are times when they labor in work that's completely irrelevant to the progress of the other processes. There are times when you are sweating. Although we are producing products at very high speed, about half of them are defective and require rework. They are trying hard, but these are examples of low work effectiveness that does not lead to added value. This is an unnecessary use of precious energy, and this does not make good use of the people's true capabilities.

The supervisor is the one who decides on how work is done. So, if the effectiveness of the work is low, it is the responsibility of management. Management, supervisors and staff must keep improving the contents of the work daily so that the people can be highly effective in their work.

Next, let us assume that a lot of effort has been continuously made towards improvement, and as a result, the *effectiveness* of the production went up, and we started making relatively more items than we used to. In cases like this, we say that the efficiency of the work went up.

Efficiency is used to compare yield, as in how many items, and with how many people, we made within a unit of time (for example, per hour). In this case, our reference (standard) is often something derived from the achievements we made so far, like the previous month or the average of the past six months. It's used as in "*This month, our efficiency went up by 15% compared to the standard.*" Accordingly, unlike effectiveness, it can go above 100% at times.

We often see many ways of expressing the ratio between the *work results* and *effort* spent in a company. Other than "*Efficiency*," there are many different measures used to evaluate whether work is progressing effectively or not, such as "*operation ratio*," "*labor productivity*," "*SPH* (amount of production per unit time.)"

There are two or three points which we should particularly pay attention to when we are using such *measures* to evaluate the *results of work*.

(A) Whether you use operation ratio or SPH, raising these values is not the objective itself, your objective is cost reduction.

By ignoring all kinds of conditions and raising the operation ratio or the SPH, then, depending on the situation, we could end up increasing the cost.

For example, to raise a line's operation ratio, we can resort to methods like making each process have intermediate work in process that's enough to cover up any minor equipment failure, or keep large amounts of all kinds of product parts, and make it such that both parts as well as their setting

methods are available to avoid being affected by a stock out in a preceding process; doing things that way can probably be thought to increase the operation ratio.

Through our past 30 years of experience in work site management, we can tell that these methods often lead to increasing the costs. It is essential to keep in mind that we can use these “*measures*” effectively only if we constantly make it clear whether they coincide with our objective or not, and fully understand the conditions of each when using them.

(B) The way we regard efficiency is extremely important

When it comes to mechanical equipment, they are most effective during the machine cycle time (the uninterrupted machine operation time). We need to clearly find out where the machines we are currently employing stand in terms of capacity, and we should keep in mind that we can increase their capacities if we ever need to.

In the case of people, it is necessary to separate movement and work. Within the cycle time, even if the people are continuously moving, we can always divide that into things we need, and things we don’t. The former is a movement that increases the value-added (in the TPS, we refer to this as “*work*” to differentiate it from simply moving), as well as the indispensable movements that are connected to it, while the latter includes all other movements, that’s to say, ones which bring no harm if they are omitted. This is what we call “*unnecessary activity*.” Do not think of this as the ability of a person, while unnecessary activity is still included in the process.

(C) It is important to sufficiently consider the sense of time, as in “*being faster*.”

Performing work faster only bears meaning once it allows for handling even more processes; hence performing work with fewer people overall. Making products faster, thus making more, can lead to an increase in efficiency. However, depending on the situation, this could result in losses. As we discussed in the beginning, efficiency is just another “*measure*.” We must only use it while checking it against our objective.

Objective is Cost Reduction

As mentioned in the previous section, the objective of improving efficiency is in reducing costs. If increasing the efficiency became the objective, there is a risk of making management mistakes. When high efficiency coincides with low cost, there will be a meaning in improving capabilities. For this reason, you must always have your eyes focused on having those two things meet, and then act.

For example, we often see work sites that set raising the SPH (amount of production per unit time) as their management target.

Note: This is not a ratio but since it’s used as a scale to measure productivity, it could be seen from the same point of view.

A production line has been using a method of putting a blackboard at the end of the line, and writing down their production output every hour. If the work site continues to do this, before anyone would realize, they will end up falling for the delusion of increasing SPH itself becoming their objective. To increase SPH, you must keep setup changes to a minimum, and to produce using large lots.

Even if today's share has been already produced, if there's still time left, they continue producing the next day's share, and that of the day after it. Certainly, the SPH would go up; but does this really lead to cost reduction? Those doing the work believe that their high efficiency is bringing profits; but, this would only lead to increasing the mountains of inventory between them and the subsequent production process.

In this case, the first condition for this line is to produce using lots that are as small as possible that correspond to the intake of the subsequent process, and to produce the required amount only. Only if you increase the SPH under these conditions, will result in cost reduction. While removing these conditions and simply increasing the SPH, will affect the whole production plant negatively instead.

This means that high efficiency and low cost are not always equal. In relation to this concept, we differentiate between the operation ratio/operational availability in the TPS in the following manner:

Operation Ratio and Operational Availability

Operation ratio is the actual production results of a machine at the present point in time, in relation to the ability of this machine when it's fully operating.

Consequently, it's natural for the monthly produced number of units to fluctuate, since there are times when it decreases if sales are deteriorating, and on the contrary, there are times when you must reach 120% or even 130%, through long overtime hours, or extra work shifts, when there's high demand. (The full regular time operation equal to 100.) Therefore, raising the operation ratio when not necessary is a big loss.

On the other hand, operational availability means to be in a state of being able to operate whenever you want to do so; and having 100% operational availability is the ideal. For this purpose, preventive maintenance must be carried out reliably, and you must aim to shorten the setup and changeover times.

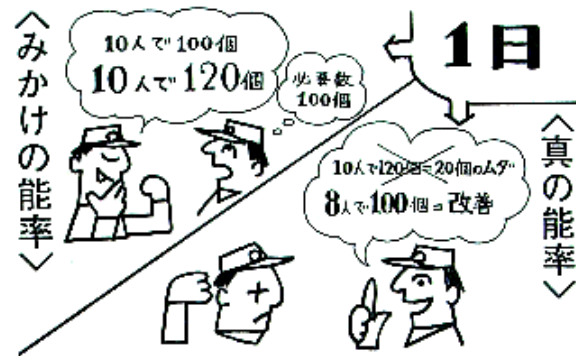
You must strictly differentiate between operation ratio and operational availability when you consider things. To understand this way of thinking, apply this on the car you own yourself, and it will be easier to comprehend. Operational availability is the ratio that expresses the state of the car as it runs in good condition when you ride it whenever you want to; so, it's desirable for this ratio to be 100%. On the other hand, the operation ratio is the ratio of how much time you drive the car each day. Since everyone drive their cars only when they need to do so, we can't say that

100% operation ratio is the ideal. Instead, if you drive it around from morning to night without any purpose, this will only lead to increasing your gas and oil consumption more and more, and the chances of having an accident will also increase; so, it will be a loss.

We can say the same about the production amount and number of personnel. Which means, if you express the relationship of how much to produce using how many people in the terms of efficiency, which of the options will directly lead to cost reduction. Here, too, we must make it clear that increasing efficiency and cost reduction are not necessarily equivalent to each other.

True efficiency and apparent (calculated) efficiency

Let us consider the following example: The line uses 10 people to produce 100 items per day. By applying improvements on this line, the capacity went up so that it can produce 120 items per day with 10 people. This is a 20% increase in efficiency. The implementation of this improvement came at the time when production quantity was needed to increase to 120 items per day.



In this case, we managed to avoid increasing the number of additional people needed by two, while being able to raise the number of items produced. Thus, it's an improvement that leads to profit.

What would happen if the production requirements stayed as 100 items per day, or even lower to 90 items per day? When this happens, if we make 120 items per day just because our efficiency was improved, we would have 20-30 extra items per day. Not only will we be consuming the raw material costs and labor costs before we needed to, but the pallets, storage places, and labor required to manage this inventory would also go up, which would be an extra cost for the company. This would be an improvement that doesn't improve the company's results, rather, it would be a change for the worse.

What should we do to improve efficiency in a way that leads to profit, even if the required production amount doesn't change or if it decreases?

In this case, we must apply the improvement on the production process to produce 100 items per day with 8 people (if the required number is 90 items per day, then we will do it with 7 people.) This will improve efficiency as well as reduce costs.

We have two ways to make the same efficiency improvement. One way to build 120 items per day with 10 people and another way to build 100 items per day with 8 people. Both methods represent about 20% increase in efficiency, but we must not forget that we are choosing the method based on how many items need to be produced.

These two methods, which may look similar according to calculations, show a great difference when they are implemented in a line. The method of increasing efficiency by raising the number of units is relatively simple, and most supervisors can carry it out, but the method of increasing efficiency by decreasing the number of personnel is several times more difficult. When we think we should increase efficiency no matter what, we tend to ignore the required amount, and pick the former method.

In many work sites, including those in other companies, we often see inventory piling up behind a line like a mountain because goods are produced at high speeds under the enthusiastic cheers of raising efficiency; however, we believe that one reason for this is forgetting that the required amount is already a premise, and the other reason is that raising the efficiency by increasing the amount is easier.

If we attempt to increase efficiency with this method when production should be reduced, we will be having an increase in expenses on one hand when sales are decreasing on the other hand, and in the worst-case scenario, this could be fatal for a company (The blow is especially painful in small to medium enterprises).

In TPS, the method of raising efficiency by increasing the amount, when the required amount is unchanged, or when production is to be reduced, is referred to as the “*apparent efficiency* (in other words, an increase in efficiency that is only based on calculations,)” and it is considered as something that we must never attempt.

When we must increase efficiency by reducing labor, then we must take that challenge, no matter how hard it is. If we become capable of doing things this way, increasing efficiency during times when production is to be increased would be incredibly easy. In this sense, we can say that periods of reduced production are a chance for the management, supervisors, and staff to grow stronger.

“Production volume = required number” is a major premise

Now that we have discussed it, we must consider what the “required amount” is.

As we have explained in the previous section, there are two methods for increasing efficiency. One is to increase the production quantity, while the other is to reduce the number of personnel.

In actual lines, if it turns out that choosing either of them works, then most lines would go for increasing the production quantity. This method is less troublesome, and only little resistance will be shown towards it.

While this method is appropriate in lines which increase the production quantity month by month, or those which have had their hands full with overtime work because of their lack in capacity,

applying it to any other case would bring about unnecessary activity of overproduction, and would lead to an increase in the cost as we have previously explained.

We have said this over and over, our objective is only cost reduction, so when it comes to increasing efficiency, the method that complies with this objective must be chosen.

To eliminate the unnecessary activity of overproduction and reduce the cost, the production quantity and the required amount must be equal. What do we use to decide on the required amount? That would be the demand. In other words, it's decided from the trends in the market. For a work site, it's something to be taken as is, and they must know that they can't just increase or decrease the amount on their own.

The increase in efficiency that's to be carried out in the production site, must regard the required quantity as a premise, and achieve this increase within its boundaries.

When using efficiency as a scale, its foundational concepts must be as described above. Once we have sorted that out, we will be able to avoid the emergence of the unnecessary activity of overproduction. Then it will turn out that *"Increase efficiency = Cost reduction."*

We would like to have a more detailed look at improvement in situations when the required amount hasn't changed from that before improvement, or when it has decreased.

In these cases, increasing efficiency while aiming for labor reduction, will only be achieved through reducing the unnecessary activity in the operations done within the Takt time derived from the required amount, getting workers to do their jobs (work) as much as possible, or automating a part of the work done by people and getting them to do some other work instead, then summing all of that up, and carrying out the production routine with even less people.

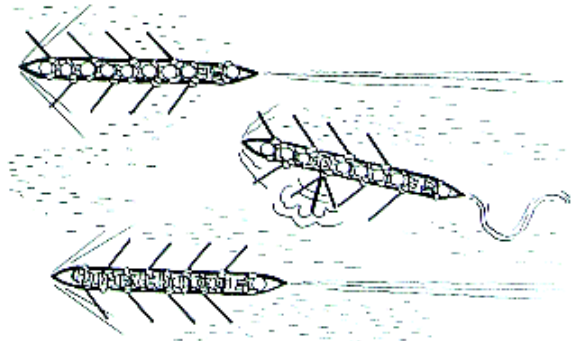
At such times, the money for mechanical equipment, jigs, tools, and transportation equipment, would have already been spent, so they should have no future effect on cost, regardless of how they are used. We should just use them according to the Takt time that's derived from the required amount, and the operation ratio should be decided accordingly.

This is as described in the previous section.

Section 2 – Individual Efficiency and Overall Efficiency

The general idea of efficiency was mentioned in the previous section. Before concretely applying this to the factory, let's consider one of the following examples.

The boat race we have in our company every fall, is a very intriguing example when we are considering efficiency. For instance, the boat can only move at a great speed once the eight oarsmen are moving in complete synchronization. The normal pitch is about 40-50 (the number of strokes per minute), and they should maintain it by following the signal. We can say that a boat race is about how good or bad the eight oarsmen are at keeping balanced.



So, what would happen if one of the eight oarsmen said, *“I’m strong, and I have the skill, so I’m going to row with a pitch of 60 strokes!”* and then he started rowing faster only by himself? His original intention was to increase speed, but that resulted in their balance crumbling, and they became slower instead.

In other words, the total force of the eight people which is moving the boat forward went up a little, but the effectiveness dropped. Not to mention, if the eight oarsmen were to row randomly at their full power, the boat wouldn’t move forward. Therefore, in a boat race, the most effective method would be for all members to maintain the pitch thoroughly; and having an individual row faster is as bad as having an individual rowing slower.

The way production works in the workplace is very similar to a boat race. The pitch is the production takt time, and by maintaining it while producing, the overall efficiency would go up.

If one operator or one line were to ignore the takt time and increased their production speed, the calculated individual efficiency may go up, but the overall efficiency would drop.

Let's consider this by concretely applying it to the factory. The smallest building blocks of a factory consist of the individual people and each piece of equipment. When they are gathered around a part or a process, a production line is formed.

Furthermore, when several lines gather together, along with the surrounding departments such as transport, inspection, and maintenance, as well as the engineers and the staff, they form a factory.

When considering efficiency with the goal of reducing costs, it is necessary to improve efficiency at all levels. That’s because every single worker and every single process throughout the whole work site are very closely connected in all directions, and there’s almost nothing that stands independently.

One person's method of working could affect somebody else, and one line's production method could often restrict the preceding and/or the subsequent processes, as well as transport operations. When discovering unnecessary activity, and pursue its cause to eliminate it we often reach the issues in the preceding, subsequent, or transport processes.

It is not just about an individual, or their own process, but it also shows us how this has a great influence on how good or bad the efficiency of the subsequent and the preceding processes would be. In this sense, even if efficiency improvement leads to cost reduction, an efficient way must be done at all levels.

In this section, we will briefly discuss increasing the efficiency on each level.

Efficiency of the Work of Each Individual

The important point at this level is the idea of "*value added ratio*". The ratio of work which increases the value-added, throughout all the work the person performs; which means that the greater it is, the higher their efficiency would be.

For example, work such as trans-shipping goods, unwrapping, and folding wrapping paper over, is originally unnecessary work. By omitting those, and only picking up work that's essential, the ratio of the work that increases the value added within one cycle of work would go up.

As related to this section, TPS has the following idea.

The Difference between Work and Activity

In Toyota, it could be said that we were more active than the Westerners. And yet, we were behind when it came to labor productivity.

We may have been more active, but we were producing less value adding work. The effectiveness of our work was lacking.

For example, we are working (working) in production activities for 7 hours 30 minutes a day, but there's too much time that's not considered work within the production activities.



In TPS, only when processes progress, and the job is fulfilled (when the value-added goes up), can we say that work has been done. Work is movement coupled with human knowledge, and we can also say that it's when the people are provided with their supervisor's knowledge.

On the contrary, we don't refer to operations which bring no harm when they are omitted (ones that don't increase the value-added) as work, but rather, as movement.

Labor Density

It is important for everyone to make efforts to convert movements to work, and conversion will lead to a reduction in labor.

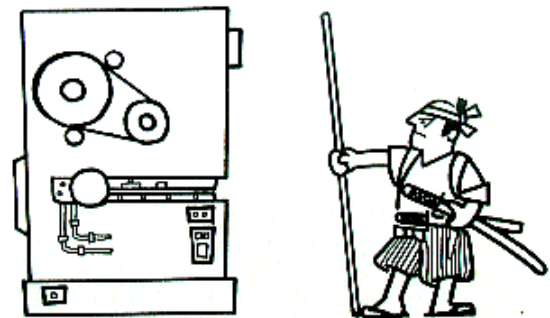
In TPS, we perceive labor density as “*work labor divided by movement labor*”, and when this ratio is high, it means that much of the worker’s energy is allocated into an important job, which would be something favorable.

This is an easy-to-understand example as to what to emphasize on reducing labor, so I will open it here.

Automation and Autonomation¹

Automated machines move even without people. But what is important as well as moving is to stop in case of abnormalities.

A machine that doesn’t stop will keep producing defective products, and there’s a high possibility of it breaking down because of working in vain when there’s nothing inside or because of foreign matter contamination. Therefore, we must assign someone to watch over it no matter what. And this person doesn’t do any kind of job that increases the value-added.



In TPS, we clearly differentiate between a machine that only moves (automated machine), and one that can automatically stop (autonomated machine) when it detects an abnormal condition.

This is the spirit of Jidoka which has been traditionally accepted since Sakichi Toyoda's invention of self-stopping loom.

When you spend so much money to achieve autonomation, there must be a befitting return (labor reduction). It would be a problem if we end up with automation instead.

We must apply an improvement after the other, and make effort so that we can bring it into the realm of autonomation.

Autonomation (自働化) includes human knowledge (a human radical 亻), while automation (自動化) includes no human (自), so it needs someone to watch over it, so we can remember them like that.

We must carry out autonomation, even in processes which contain a lot of manual labor, such as final assembly conveyors, by making an interpretation of these concepts.

¹ Also known as ‘Jidoka’.

In this case, the “character for ‘self 自’” refers to the person, and it means allowing the person themselves to stop the conveyor if they ever think *“this isn’t right!”* or *“this is a defective product!”* throughout the work they are performing.

If we were to put it in an extreme way, every single person would be carrying a line stop switch, and the moment they think something is odd, little as it may be, they would stop the line immediately.

Producing defective products or products that don’t pass that standards, won’t be considered as doing one’s job. It won’t be considered as work, and this, as well, is one more viewpoint of automation.

Efficiency of Work in the Production Line

There are two important things at this level:

Balance between Processes²

In other words, this means production according to the takt time and avoiding overproduction.

When some workers set their own high efficiency as their aim and produce quickly, products will pile up in front of the workers of the subsequent production stage, and the work of sorting these out will increase. They might start to have a feeling of impatience, and mistakes will increase. If you look at the whole production the line, the efficiency will decrease instead.

Generally, when there’s work that’s done by several people in order, there’s always a production stage that is considered the bottleneck, and it represents the capability of the line.

To increase the efficiency of the whole line, you must help the people of this bottleneck production process; however, since those who can work fast keep producing, items keep piling up in front of this bottleneck process. This will hold everyone back instead, and will cause efficiency to decrease.

To prevent this, you must clearly decide the standard operating procedures and accordingly, you must make everyone abide by the standard in-process stock, the takt time, and the work sequence, and since by doing that there will be waiting, you must do compositional changes and keep the balance.

For steps that involve working in a row like presses, it’s essential to modify them such that the conveyor stops when a part enters within the reach of the worker’s hand.

² Flow Principle #3 – Balance work content.

Mutual Cooperation (relay races of swimming and track)

In the above-mentioned manner, it is often that it is impossible to completely balance the people. It's unavoidable to have little unbalance in the work of a production line that has several people.

In this case, it's possible to increase the efficiency of the whole line, by having those who finished earlier aid those who are still behind.

Even if you have already allotted the production processes, it's desirable for the boundaries between them to be flexible to allow for mutual cooperation.

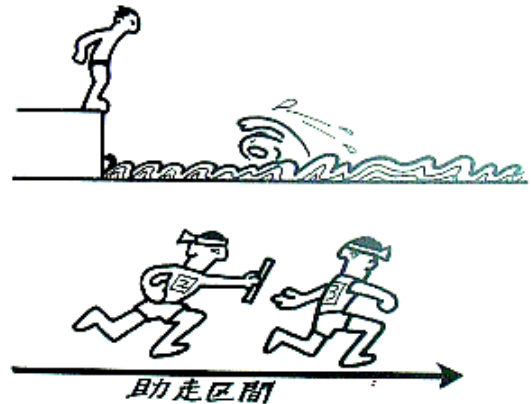
For example, if the previous process is completed a little earlier, you should devise a way to put the product in the jig of the next first step.

As for this way of thinking, please refer to the way of swimming and Baton Pass of the Relay Race as a reference.

In the relay in swimming, after the first swimmer touches the pool side, the second swimmer jumps in. In track, there is a 20-meter baton pass zone, and handing over the baton anywhere inside this zone is acceptable.

In swimming the slower individual and the faster individual must do 100m each, while in track, it's possible to use the "strategy" of having the faster individual run 120m, while the slower one runs 80m.

The work done in the actual site must be a relay of track. It's desirable to create a baton pass zone so that the supervisor can have a "strategy" that increases the efficiency.



Efficiency of the Whole Factory

Sometimes we come across examples where the efficiency of the factory doesn't increase in companies that put a lot of effort in increasing the efficiency of every individual.

Despite the manual labor of the people being splendidly fast, the inventory is piling up in the factories, and this requires many people to sort out everything, and as an overall result, let alone cost reduction, this will lead to the cost going up.

This is because productivity only applies to the work speed of the individuals, while ignoring the overall condition. The efficiency of people, same as the efficiency of the line, and the efficiency of the factory, all affect the cost greatly. It is essential to give the most priority to increasing efficiency on the level of the entire factory. To improve the efficiency of the factory, certain rules are necessary.

This is not something that can be done by one supervisor or one staff member alone. For everyone to move in the same direction, standards of thinking become necessary. The standard is "*leveling production*" in TPS.

In the case of teamwork, it must be a track relay. The supervisor is advised to create a baton touch zone so that "*strategy*" can be created to improve the efficiency of the team.

Production Leveling

For a production site, the more the flow of products varies, the more unnecessary activity occurs. That's because equipment, people, inventory, and everything else must meet the necessary production peak and be prepared accordingly (capacity planning to meet peak demand).

If it is built in such a way as to disturb the pace in any step, unnecessary activity occurs and the efficiency decreases, which is not only a problem, but it has a serious influence on the preceding and subsequent process, in particular the preceding ones.

To explain this in an easy-to-understand example, what would happen if a final assembly line was concerned only with the efficiency of its own process, and decided to go with producing sedans today, hardtops tomorrow, and vans the day after that?

The production line of preceding process that's working on sedans has work to do today, but the line working on the hardtop or van parts, have no work until tomorrow, or the day after respectively.

Having no other choices left, they reschedule their production plan so that they have work to do day by day using their inventory, but even if they do so, they will at least need inventory that equals 3-4 times of what they would have needed if the assembly line assembled the average amount day by day.

With such a thing, it is unexpected that efficiency improvement etc. Leveling production is this meaning. Leveling includes reducing the variance of types of goods and eliminating variations in quantity, as well as reducing the lot of transportation and eliminating variability.

Small Lot Production

Manufacturing processes of automobiles have many processes ranging from casting, forging, rough shapes such as presses, machining, welding, intermediate processing, painting and assembly.

To what extent leveling can be accomplished, including all these steps, is just how far we can improve the efficiency of the whole factory.

For instance, metal-casting (die casting), forging, presses, and heat treatment are process that leave no choice but to be done using lot production, and for a long time, there has been many

cases where they have been treated as a separate matter, despite being processes that need to receive the most efforts when leveling.

In other words, lot production components are excessively fixated on SPH (amount of production per unit time) and the setup times, and they think that having excessively large lots leads to increasing efficiency.

As explained earlier, enlarging the lot confuses the process before and after (in this case not the lot production), it generates excessive inventory and a lot of unnecessary activity. Consequently, regarding lot production components, it is important to make the lot small and make efforts to level production.

Section 3 – Increasing Efficiency by Eliminating Unnecessary Activity

Waste of Capacity

If we were to sum up what we have discussed in sections 1 and 2, we will end up with the following:

(A) Improving efficiency is meaningful only when it leads to cost reduction.

To achieve that, we must move in the direction of “how do we make only what we need with few people?”

(B) When we regard efficiency in terms of every single person, as well as the lines at which they gather, and the whole factory which is centered on these lines, then we must proceed with ideas and viewpoints aiming to increase efficiency on each of these levels; the overall results must go up as well.

Next, we would like to consider what we should exactly do in a more easy-to-understand manner.

Let's take an example for a line. In this line, 10 people make 100 products a day. Based on this current situation, the capacity of this line is 100 per day, and the productivity per person would be 10 items a day. If you look closely at this line and the people, you will find overproduction or waiting time, or we can find variations depending on time and day.

Assume we applied improvement to these, and reduced 2 people's worth of labor. In this case, the capacity would seem like it has gone up, compared to how it used to be. (If we don't remove the two people, we will be able to produce 125 items/day, so it would seem like capacity has gone up by an amount of 25 items).

The truth is, we had the capacity to produce 125 items/day from before. But it's just that the 25 items worth of capacity was wasted by things like unneeded operations or overproduction. It is appropriate to think that we were able to make only 100 items a day because so much unnecessary activity was included.

Whether we're looking at it in terms of every single person or the whole line, if we consider what really needs to be done as their jobs, and anything other than these as unnecessary activity, we would have the following relationship:

$$\text{Current Capacity} = \text{Useful Work} + \text{Unnecessary Activity}$$

The more we get rid of unnecessary activity, and increase their jobs in an equivalent amount, the more we will approach the “true capacity” (100% value adding work as the true capacity). When we do that, we will make more products with the same line configuration.

We must not make more than the required amount. Reduce the number of people to reduce excess capacity to make exactly what you need. This is the same when it comes to each person, to the production line, or for the whole factory.

For each person, we focus on the work that adds value and the work that absolutely is necessary in relation to it, and avoid other unnecessary activity. On the line, we allocate highly efficient work in order, and eliminate labor fractions (fractional person labor) by improvements. When it comes to the whole factory, we try to establish an overall balance around such lines, make the lots as small as possible, and prevent the emergence of extra transportation costs by carrying out production leveling.

Simultaneously, we apply improvements that prevents the increase in the labor for investigations and readjustments due to frequent occurrences of defective items.

If such actions are taken, unnecessary activities will disappear gradually, so the people, the lines, and the factories, would be able to do an amount of their necessary jobs. In other words, their hidden capacities would come to light. If we carry out labor reduction now, the efficiency will go up as explained in Chapter 1, Section 3.

When we start thinking like this, we can see that raising efficiency means saving unnecessary activity. A thorough exclusion of unnecessary activity, this is the most basic thing that we must do.

Types of Unnecessary Activities

Within the TPS labor reduction routine, there are four categories of unnecessary activities we want to remove:³

- (1) unnecessary activity of overproduction
- (2) unnecessary activity of waiting time.
- (3) unnecessary activity of transport.
- (4) unnecessary activity in processing itself.

We will explain them one by one below.

Unnecessary Activity of Overproduction

It is unnecessary activity that occurs to make extra things. The unnecessary activity of this overburden is often caused by excessive progress of work in normal cases. Even though it is time that I must wait normally, I start the next task.

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= 造りすぎ



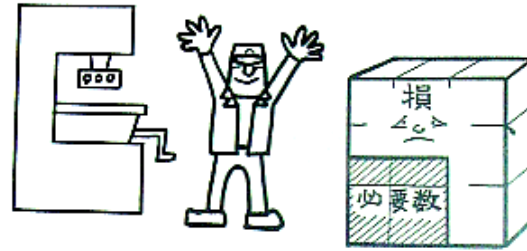
= 手待ち



= 運搬

³ These four categories were further expanded and subdivided into what is commonly referred to today as the '7-Wastes'. The objective of defining them was to make it easier for people to see where improvements needed to be made.

For example, the work of piling up and unloading that's seen between one person and the other, or in the connection of one process and the other, and these lead to the causes that increase unnecessary activity. These are the cause of unnecessary effort. Unnecessary overwork such as additional machinery, advanced use of materials, unnecessary replenishment of workers, interest burdens, wide warehousing and transportation expenses is not limited to self-process, but also causes other unnecessary activity. Even illusion is made as if it is working.



There are many causes for the unnecessary activity of overproduction other than this. We can think of the following examples:

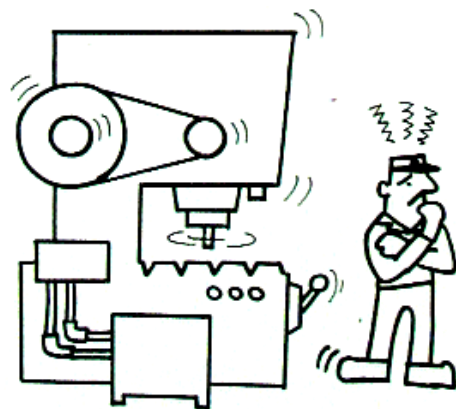
- (A) When you allow for surplus in work, or when there are waiting hours, so as a result, work progresses more than required in this time.
- (B) When production is accumulated just because there's still room for this regarding the capacities of the machinery (equipment).
- (C) When the overall efficiency is ignored because of running after increasing the SPH (amount of production per unit time).
- (D) When you adopt the false idea of *"This machine is expensive, so if we don't increase its operation ratio, we won't be able to compensate for it."*

Unnecessary Activity of Waiting

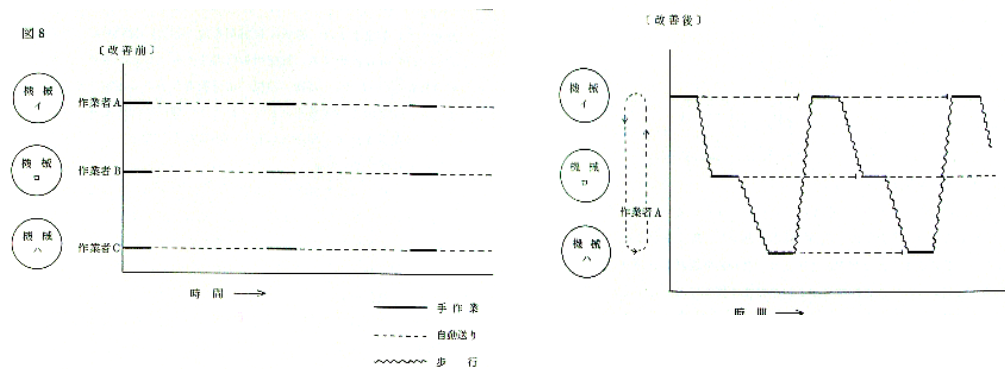
When machines are processing automatically, people are just standing and watching machines, even if they want to work, the machine is running, and they cannot leave. It is about unnecessary inactivity that occurs.

For example, before doing improvements, one person has been assigned to each one of the machines (I), (II), and (III).

In such production processes, while the machines are doing the work, the people just stand idly, and can't do work even if they want to do so. The unnecessary activity of waiting time is taking place.



To eliminate unnecessary waiting, we decided to have one worker do the loading of all three machines in sequence.



Which means, operator A will install the materials for the machine, and after pushing the start button, they will move to machine (II). Then they set the materials for machine (II), and start it. After that they will move to machine (III), and set the materials in the same manner as machines (I) and (II), start the machine, and then return to machine (I). When operator A returns to machine (I), machine (I) will have finished its work, so operator A will be able to start working immediately, waiting time doesn't occur.

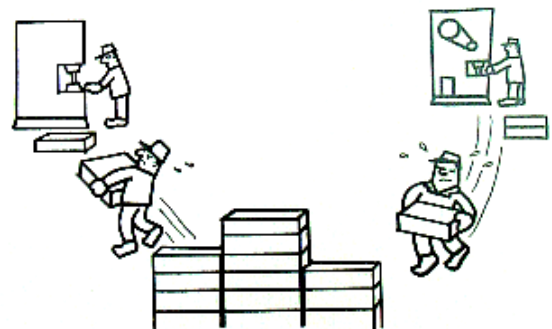
By eliminating unnecessary activity of waiting, two people can be reduced. Movement without which the “work” can still be done, such as lost motion, should be included in that. (Combining work will be discussed in detail in Chapter 3 the standard operating procedures)

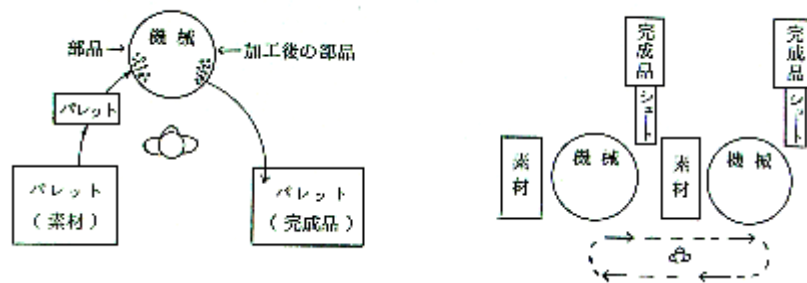
Unnecessary Activity of Transportation

The unnecessary activity of transport is an unnecessary transport distance, temporary placement, for trans-shipment and movement. It is about unnecessary activity that occurs.

To eliminate unnecessary activity of transportation,
 (A) You must keep the transport distance as short as possible.
 (B) Eliminate temporary placement.

As an example, parts were transferred from a large pallet to a small pallet, and furthermore temporarily placed on the machine (on the bed of the lathe) for temporary placement, but the pallet. By eliminating these temporary placements (eliminating unnecessary activity of transportation), we improved one machine so that one person can operate two machines.





Also, from shipping warehouse to factory, from factory to machine side, from machine side to operator's hands have been repeatedly transferred and moved, but these can also be said to be unnecessary activity of transportation.

Unnecessary Activity in the Process

For example, the work is done while the jigs are supported by the left hand, and for that reason the manufacturing of goods isn't progressing smoothly, and the wasted time is increased.

This is unnecessary activity of the processing itself. As an improvement in such a case, it is important to repair the jig as soon as possible.

Discovery of unnecessary activity is the first step towards increasing efficiency

We have already discussed how the TPS's central ideologies are all about elimination of unnecessary activity, and that unnecessary activity can be roughly classified into the four types in the previous section. When you consider each of these four, you can see that there are unnecessary activities that are easy to find, and some that aren't.

Few managers and supervisors let their people do unnecessary activities while being aware that it is not useful. In most cases, they regard it as necessary work, or just lack understanding.

No matter how much they think "Let's eliminate unnecessary activities!" if they don't understand where the unnecessary activity is, they won't be able to proceed with its elimination at all.

Therefore, what really matters is making unnecessary activity visible to everyone.

That is the first step to increasing efficiency. In the previous section, the easiest unnecessary activity to find among the four types would be that of waiting time.

For example, within a three-minute takt time, a minute of waiting time emerges every cycle, not only the supervisor, but the person and the senior managers can tell that there's surplus.

If the operator moves around as if they are doing work throughout this minute, it will be hard to tell (unnecessary activity of transport and processing). If the next product is processed in this minute, nobody would be able to tell whether it's an unnecessary activity or not (unnecessary

activity of overproduction). All four types of unnecessary activities need to be changed into waiting time. That way, it would be relatively easier to put together countermeasures.

To do that, we consider methods like (A) Making the operator follow the standard operating procedures thoroughly, and not letting them do anything other than them. (B) Restricting them from getting far ahead, using a production system that utilizes Kanban. (C) Specifying a work space on the conveyer, thus preventing the operator from handling items ahead of time.

We will leave the detailed explanation of these methods to another chapter, but what's important here, is that we must be fully aware that to eliminate unnecessary activity, finding it comes first.

We must be aware that we should constantly maintain the work site so that any unnecessary activity is easy to find. These can be very minor matters at times, for example, there are times when the matter in question is just a single piece of inventory. Every increase in efficiency leads to cost reduction, it's important to think about why an item was made, even if it's a single item. Sometimes it is enough to think that this will serve as a clue for improvement.

In this section, we have discussed how increasing efficiency is carried out by eliminating unnecessary activities. We believe that there are still other ways to distinguish unnecessary activity, but the focus is that we should make an effort of changing the different types of unnecessary activities into waiting time, which is easy to see. We would like to repeat and emphasize that this is the first step to increasing efficiency.